
Linear systems of equations

Discussion, solution and geometric interpretation

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Introduction

In this module we review the basic concepts of systems of linear equations (SLE), their expression using matrix notation, their discussion (i.e. study the number of solutions of the SLE) and two different methods to compute their solution: Gaussian elimination and Cramer's rule. This module also includes an introduction to geometric interpretation of SLE with three variables (that is, the 3D space).

Many economical, physical and technological phenomena could be modelled (exactly or approximately) using linear systems. For example, Leontief's models, or *input/output* tables, use SLE to describe the existing relationship between supply and demand in different sectors from a national economy. Linear systems find applications in a wide range of areas, from the balancing chemical equations to analysing the data flow in a network or in linear programming optimisation.

Objectives

The main goal of this module is to provide the students with the main concepts and mathematical rules associated with the discussion, resolution and interpretation of **systems of linear equations** (SLE) or simply linear systems.

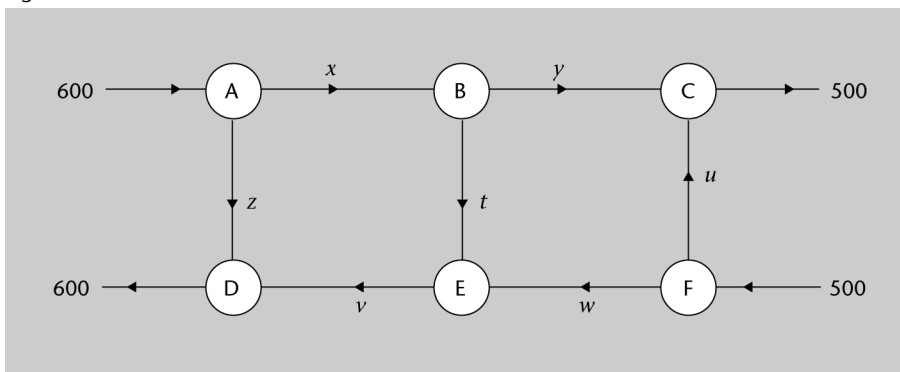
Concretely, the specific aims of this module are the following:

1. Understand the concepts related to linear systems and use matrix notation to express linear systems.
2. Calculate the rank of a matrix and use Rouché-Capelli theorem to discuss linear systems.
3. Study the characteristics of homogeneous systems of linear equations.
4. Learn to solve linear systems using Gaussian elimination and Cramer's rule.
5. Interpret linear systems from a geometric perspective.
6. Use numerical-analysis software as a way to (a) experiment with main concepts of this module and (b) automate discussion and resolution of linear systems.

1. Initial example

The schematic in Figure 1 shows the data flow, measured in mega-bytes per hour (MB/h), between six routers (nodes A-F) that constitute a subnetwork of a wide area network (WAN):

Figure 1



Let us consider that the stationary data rate present in the incoming connections to the subnetwork (through nodes A and F) is 1100MB/h and that the network outputs data (through nodes C and D) at the same rate.

Moreover, we assume that the data flow coming and going from each node is the same.

Initial example

In this setting, we ask:

- Calculate the data flow between each pair of connected nodes, i.e. calculate the unknown variables x, y, z, t, u, v, w .
- Given that $u = 300$ and $v = 100$, calculate the data rate between each pair of connected nodes.

The answers to this initial example are thoroughly explained in the *Solutions to exercises* section at the end of this module.

2. Systems of linear equations (SLE)

A system of m linear equations with n variables (SLE) can be written in the form

[illegible]

where:

- $a_{ij} \in \mathbb{R}, \forall i = 1, 2, \dots, m, \forall j = 1, 2, \dots, n$ are the **coefficients** of the system, usually known in advance.
- $x_j \in \mathbb{R}, \forall j = 1, 2, \dots, n$ are the unknown **variables** of the system.
- $b_i \in \mathbb{R}, \forall i = 1, 2, \dots, m$ are the **constants** of the system, usually known in advance.

Often, systems of linear equations are used to model real case scenarios. In those cases, both the coefficients a_{ij} and the constants b_i are known a priori and the goal is to estimate the value of the variables x_j such that all equations are jointly verified. All possible sets of variables that jointly satisfy all the equations are called **solution sets** and the process to find such set is called resolution of the system. Moreover, two systems are called **equivalent** if they have the same solution set. Even though two equivalent systems can have different number of equations, they always have the same solution set.

Example 1. Different systems of equations with its solution

a) The solution set for the following system of three linear equations

$$\begin{cases} x + y + z = 2000 \\ y + z = 1500 \\ x + y + \frac{3}{4}z = 1800 \end{cases}$$

is $\{x = 500, y = 700, z = 800\}$.

b) The solution set of the SLE of three variables

$$\begin{cases} x + y + z &= 2000 \\ y + z &= 1500 \\ 4x + 4y + 3z &= 7200 \end{cases}$$

Observation

Usually, x, y, z are the characters used to represent the variables in a SLE of 3 variables.

is $\{x = 500, y = 700, z = 800\}$ and hence, it is equivalent to the previous SLE in (a).

- c) If the relationship between the variables is not linear, the system is considered a system of nonlinear equations (SNLE). The following SNLE

$$\begin{cases} y = 5(x-1) \\ y = x^3 - x^2 \end{cases}$$

has multiple solution sets: $\{x = 1, y = 0\}$, $\{x = \sqrt{5}, y = 5\sqrt{5} - 5\}$ and $\{x = -\sqrt{5}, y = -5\sqrt{5} - 5\}$

- d) The SNLE with two exponential equations and two variables

$$\begin{cases} 2^{x+y} = 1 \\ 2^x - 2^{y+2} = 0 \end{cases}$$

has $\{x = 1, y = -1\}$ as the solution.

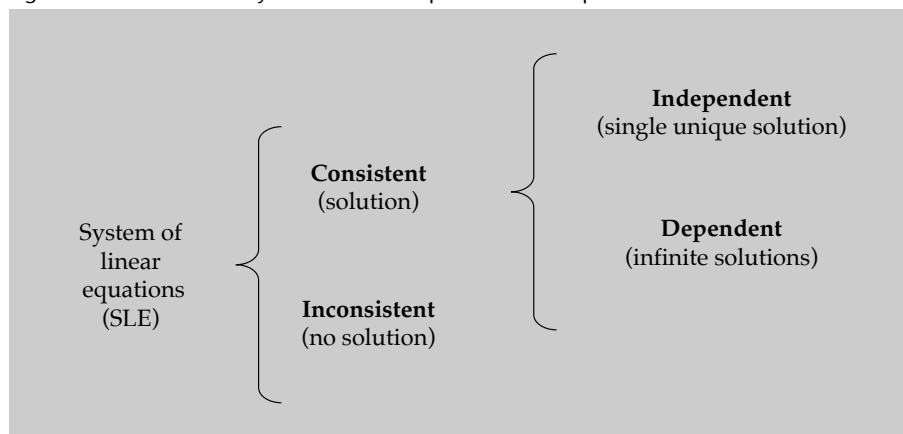
- e) The following SLE with two variables

$$\begin{cases} x + y = 1 \\ x + y = 2 \end{cases}$$

is undetermined and thus does not have any valid solution set (note that two real numbers cannot sum up to 1 and 2 at the same time).

In general, any system of equations (linear and nonlinear) is said to be **consistent** if it has at least one valid solution and **inconsistent** if it has no solution. Moreover, in the specific case of SLE, a consistent system may have a single unique solution (**independent**) and infinitely many solutions (**dependent**), as summarised in Figure 2.

Figure 2. Classification of systems of linear equations with respect the number of solutions.



3. Matrix notation

A system of linear equations (1) can be expressed using matrix notation as

$$\begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \cdot \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad (2)$$

or, more compactly,

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{B}, \quad (3)$$

where

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n}$$

is the **coefficient matrix**, $\mathbf{X} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} \in \mathbb{R}^n$ is the vector of variables and

$\mathbf{B} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \in \mathbb{R}^m$ is the constant vector.

Using the coefficients and constant terms of the SLE, we define the **augmented matrix** of coefficients

$$\mathbf{M} = (\mathbf{A}|\mathbf{B}) = \left(\begin{array}{cccc|c} a_{11} & a_{12} & \cdots & a_{1n} & b_1 \\ a_{21} & a_{22} & \cdots & a_{2n} & b_2 \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} & b_m \end{array} \right) \in \mathbb{R}^{(m+1) \times n}$$

Example 2. Matrix notation

The following system of three equations and four variables

$$\begin{cases} x - 3y + 6z - 8t = 5 \\ y + z = 15 \\ x + y - 3z + t = 8 \end{cases}$$

can be expressed using matrix notation as

$$\begin{pmatrix} 1 & -3 & 6 & -8 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & -3 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 5 \\ 15 \\ 8 \end{pmatrix}$$

Observation

The initial SLE expression can be recovered by operating the matrix multiplication.

The coefficient matrix ($\mathbf{A}$) and the augmented matrix ($\mathbf{M}$) are

$$\mathbf{A} = \begin{pmatrix} 1 & -3 & 6 & -8 \\ 0 & 1 & 1 & 0 \\ 1 & 1 & -3 & 1 \end{pmatrix}, \quad \mathbf{M} = \left(\begin{array}{cccc|c} 1 & -3 & 6 & -8 & 5 \\ 0 & 1 & 1 & 0 & 15 \\ 1 & 1 & -3 & 1 & 8 \end{array} \right).$$

4. Discussion of SLE

The study of the consistency and dependency of linear systems is known as **discussion of SLE**. These two main properties are generic to any linear system of equations and indicate the number of solutions a priori, before actually solving the system.

Following the matrix notation introduced, the *Rouché-Capelli theorem* is used for discussing linear systems.

Rouché-Capelli theorem. Given a system of m linear equations with n variables, then:

- if $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = n$, the system is **independent**,
- if $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = r < n$, the system is **dependent** and has $(n-r)$ **degrees of freedom**,
- if $\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{M})$, the system is **inconsistent**,

where $\text{rank}(\cdot)$ refers to the rank of a matrix.

Note that the theorem covers all possible cases since for any SLE the inequality $\text{rank}(\mathbf{A}) \leq \text{rank}(\mathbf{M}) \leq n$ holds.

Example 3. Discussion of linear systems using Rouché-Capelli theorem

We study the consistency of the following SLE:

$$\begin{cases} 3x + 2y = 1 \\ x + y = 0 \\ 2x - 3y = 2 \end{cases}.$$

a) We first determine the coefficient and augmented matrices

$$\mathbf{A} = \begin{pmatrix} 3 & 2 \\ 1 & 1 \\ 2 & -3 \end{pmatrix}, \quad \mathbf{M} = \left(\begin{array}{cc|c} 3 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & -3 & 2 \end{array} \right).$$

See also

See the definition and how to calculate the *rank of a matrix* in section 4 from "Linear Transformations" module.

- b) We then compute $\text{rank}(\mathbf{A}) = 2$. Since the minor $\begin{vmatrix} 3 & 2 \\ 1 & 1 \end{vmatrix} = 1 \neq 0$, and no larger minor can be found in $\mathbf{A}$, then $\text{rank}(\mathbf{A}) = 2$.
- c) Finally, we compute $\text{rank}(\mathbf{M}) = 3$. The largest non-zero minor of $\mathbf{M}$ is the matrix itself and thus $\text{rank}(\mathbf{A}) = 2$:

$$\begin{vmatrix} 3 & 2 & 1 \\ 1 & 1 & 0 \\ 2 & -3 & 2 \end{vmatrix} = (6 + 0 - 3) - (2 + 4 + 0) = -3 \neq 0.$$

- d) Conclusion: following the Rouché-Capelli theorem, since $\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{M})$, the system is inconsistent.

Example 4. Discussion of linear systems using Rouché-Capelli theorem

We study the consistency of the following SLE:

$$\begin{cases} 2x + y = 4 \\ x + z = 4 \\ 3x + y + z = 8 \end{cases}$$

- a) We first determine the coefficient and augmented matrices

$$\mathbf{A} = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 0 & 1 \\ 3 & 1 & 1 \end{pmatrix}, \quad \mathbf{M} = \left(\begin{array}{ccc|c} 2 & 1 & 0 & 4 \\ 1 & 0 & 1 & 4 \\ 3 & 1 & 1 & 8 \end{array} \right).$$

- b) We then compute $\text{rank}(\mathbf{A}) = 2$, since $|\mathbf{A}| = 0$ and the minor $\begin{vmatrix} 2 & 1 \\ 1 & 0 \end{vmatrix} = -1 \neq 0$.
- c) Finally, we compute $\text{rank}(\mathbf{M}) = 2$. All 3×3 minors are zero (note that the last row is the sum of the first two) and we can find the same 2×2 minors as in $\mathbf{A}$.
- d) Conclusion: since $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 2 < 3$, the system is consistent-dependent with one degree of freedom.

Note that, in this case, the last equation is the sum of the first two. Hence, in practice we effectively have a system of two equations and three variables whose solution set can be easily found by removing the last equation:

$$\begin{cases} 2x + y = 4 \\ x + z = 4 \end{cases} \quad \text{i.e.,} \quad \begin{cases} x + z = 4 \\ 2x + y = 4 \end{cases}$$

Therefore, since we have one degree of freedom, we could express two variables as a function of the third one (referred to as *free variable*).

$$\begin{cases} x = 4 - z \\ y = -4 + 2z \\ z = z \end{cases} \quad \text{or, parametric,} \quad z = \begin{cases} x = 4 - t \\ y = -4 + 2t \\ z = t \end{cases}, \quad t \in \mathbb{R}$$

Example 5. Discussion of parametric systems

We study the following SLE with respect all possible values of the k parameter ($k \in \mathbb{R}$)

$$\begin{cases} x + y + z = 1 \\ ky + kz = 2 \\ kx + ky + z = 1 \end{cases}$$

The coefficient and augmented matrices are

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & k & k \\ k & k & 1 \end{pmatrix}, \quad \mathbf{M} = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & k & k & 2 \\ k & k & 1 & 1 \end{array} \right).$$

Since it is a system of 3 linear equations with 3 variables, the rank of system matrices is less or equal than 3. We first study the case of $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 3$, for which the system is independent. In this scenario, the determinant of $\mathbf{A}$ and at least one 3x3 minor of $\mathbf{M}$ need to be different than zero. Noting that $\mathbf{A}$ is a 3x3 minor of $\mathbf{M}$, then $|\mathbf{A}| \neq 0$ is the only condition for the system to be independent.

$$|\mathbf{A}| = (k + k^2 + 0) - (k^2 + k^2 + 0) = k - k^2 = k(1 - k)$$

Hence, $|\mathbf{A}| \neq 0 \iff k \in \mathbb{R} \setminus \{0, 1\}$ and the system is independent for any real value of k except for 0 and 1. Secondly, we study the specific cases of $k = 0$ and $k = 1$, for which $\text{rank}(\mathbf{A}) < 3$ and the system would be dependent or inconsistent depending on the $\text{rank}(\mathbf{M})$:

- Case $k = 0$: the coefficients and augmented matrices are:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \quad \mathbf{M} = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 1 & 1 \end{array} \right).$$

As we already know from the previous calculation, $|\mathbf{A}| = 0$. Moreover, the $\text{rank}(\mathbf{M}) = 3$, since we could find a non-zero 3x3 minor:

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 0 & 2 \\ 0 & 1 & 1 \end{vmatrix} = (0 + 0 + 0) - (0 + 0 + 2) = -2 \neq 0.$$

In this case, the system is inconsistent ($\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{M}) = 3$).

- Case $k = 1$: the coefficients and augmented matrices are:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix}, \quad \mathbf{M} = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{array} \right).$$

As we already know from the previous calculation, $|\mathbf{A}| = 0$ and we need to compute the $\text{rank}(\mathbf{M})$. Here we have two 3×3 minors, one corresponding to the coefficient matrix and the other including the constant terms. Since we already know that the first is 0, we calculate the second:

$$\begin{vmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 1 & 1 & 1 \end{vmatrix} = (1 + 0 + 2) - (1 + 0 + 2) = 0.$$

Therefore, $\text{rank}(\mathbf{M}) < 3$. We then proceed to calculate the $\text{rank}(\mathbf{A})$ by computing 2×2 minors. If we find a non-zero minor in $\mathbf{A}$, then $\text{rank}(\mathbf{A}) = 2$. For example, selecting the minor from the first two columns and rows,

$$\begin{vmatrix} 1 & 1 \\ 0 & 1 \end{vmatrix} = 1 - 0 = 1 \neq 0,$$

we find that $\text{rank}(\mathbf{A}) = 2$. In addition, we know that $\text{rank}(\mathbf{M}) = 2$ because the same minor can be found in $\mathbf{M}$. Hence, we conclude that the system is consistent and dependent with one degree of freedom ($\text{dof} = n - \text{rank}(\mathbf{A}) = 3 - 2 = 1$).

In summary,

- For $k \neq 0, 1$, the system is consistent and independent.
- For $k = 0$, the system is inconsistent.
- For $k = 1$ the system is consistent and dependent, with 1 degree of freedom.

5. Homogeneous linear systems

A system of linear equations is said to be **homogeneous** if all constant terms are zero, i.e., the SLE can be written as

$$\begin{cases} a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n = 0 \\ a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n = 0 \\ \vdots \\ a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n = 0 \end{cases} \quad (4)$$

or, using matrix notation as

$$\mathbf{A} \cdot \mathbf{X} = \mathbf{0}, \quad (5)$$

where $\mathbf{A}$ is the coefficient matrix, $\mathbf{X}$ is the vector of variables and

$$\mathbf{0} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \in \mathbb{R}^n$$

Note: the augmented matrix ($\mathbf{M}$) of any homogeneous SLE is build by concatenating the zero vector to the coefficient matrix $\mathbf{A}$, which means that $\text{rank}(\mathbf{M}) = \text{rank}(\mathbf{A})$. Hence, following the Rouché-Capelli theorem, we conclude that **homogeneous SLE are always consistent** and that they always have at least one solution. Such solution is called **trivial solution**, namely, $\mathbf{x} = \mathbf{0} \rightarrow x_1 = x_2 = \cdots = x_n = 0$.

The important question is then whether the SLE has a unique solution (trivial) or there also exist multiple nontrivial solutions. By exploring the rank of matrix $\mathbf{A}$, if $\text{rank}(\mathbf{A}) = n$ then the system would be independent with a single solution (the trivial) and if $\text{rank}(\mathbf{A}) < n$ the system would be dependent with infinitely many solutions — among which the trivial.

Example 6. Discussion of an homogeneous SLE

Let's discuss the following SLE:

$$\begin{cases} x + y + z = 0 \\ x - y + z = 0 \\ -x + 2y + z = 0 \end{cases}.$$

Note that

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 2 & 1 \end{pmatrix} \quad \text{and} \quad |\mathbf{A}| = (-1 - 1 + 2) - (1 + 2 + 1) = -4 \neq 0.$$

Consequently, $\text{rank}(\mathbf{A}) = 3 = n$ and the only solution of the system is the trivial.

With this example, we show that all solutions of any homogeneous SLE, $\mathbf{AX} = \mathbf{0}$, with n variables build a subspace in $\mathbb{R}^n$. For the system to have nontrivial solutions, there should be at least one free variable (or degree of freedom).

6. Gaussian elimination algorithm

The **Gaussian elimination** or row reduction algorithm consists of a sequence of multiple steps operating on columns/rows of the augmented matrix, M , such that it produces a new matrix, E , with the following characteristics:

- 1) Matrices E and M represent equivalent systems of equations
- 2) Matrix E is in row echelon ("steplike") form:

$$E = \left(\begin{array}{cccc|c} e_{11} & e_{12} & \cdots & e_{1m} & \cdots & e_{1n} & d_1 \\ 0 & e_{22} & \cdots & e_{2m} & \cdots & e_{2n} & d_2 \\ \vdots & \vdots & \ddots & \vdots & \ddots & \vdots & \vdots \\ 0 & 0 & 0 & e_{mm} & \cdots & e_{mn} & d_m \end{array} \right). \quad (6)$$

In other words, $e_{ij} = 0, \forall i > j$

The basic operations that can be performed at each step are:

- Transpose two rows.
- Transpose two columns. In this case one needs to be careful as the variable order is swapped.
- Scale a single row by multiplying with a non-zero scalar value.
- Summation of a row with a scaled version of another row or with a linear combination of several rows.
- Remove a row containing only zeros.

Once the matrix E is obtained, a new associated SLE is defined with E being the augmented matrix and possibly containing some variable swapping due to column transposition. Now, an iterative process is carried out in order from the last nontrivial equation (or row) to the first one. It consists of expressing a single variable of choice as a function of the rest. Then, in the immediate equation from above, the variable is substituted with its calculated value (i.e. as a function of other variables) and another variable is isolated as a function of the rest. If we keep doing this until the first row, we would get an expression for each variable.

Example 7. Using Gaussian elimination to solve a linear system

Consider the following SLE:

$$\begin{cases} x + 3y - 2t = -1 \\ 2x + 6y + z + t = -12 \\ 3x - y + z - t = 7 \\ 2x + y + z + t = -2 \end{cases}$$

From the augmented coefficient matrix $\mathbf{M}$, we perform a set of sequential steps towards the matrix $\mathbf{E}$ in echelon form

$$\begin{aligned} \mathbf{M} &= \left(\begin{array}{cccc|c} \textcircled{1} & 3 & 0 & -2 & -1 \\ 2 & 6 & 1 & 1 & -12 \\ 3 & -1 & 1 & -1 & 7 \\ 2 & 1 & 1 & 1 & -2 \end{array} \right) \xrightarrow{(1)} \left(\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -1 \\ \boxed{0} & 0 & 1 & 5 & -10 \\ 0 & -10 & 1 & 5 & 10 \\ 0 & -5 & 1 & 5 & 0 \end{array} \right) \xrightarrow{(2)} \\ &\xrightarrow{(2)} \left(\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -1 \\ \boxed{0} & -5 & 1 & 5 & 0 \\ 0 & -10 & 1 & 5 & 10 \\ \boxed{0} & 0 & 1 & 5 & -10 \end{array} \right) \xrightarrow{(3)} \left(\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -1 \\ \boxed{0} & 5 & -1 & -5 & 0 \\ 0 & -10 & 1 & 5 & 10 \\ 0 & 0 & 1 & 5 & -10 \end{array} \right) \xrightarrow{(4)} \\ &\xrightarrow{(4)} \left(\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -1 \\ 0 & \textcircled{5} & -1 & -5 & 0 \\ \boxed{0} & 0 & -1 & -5 & 10 \\ 0 & 0 & 1 & 5 & -10 \end{array} \right) \xrightarrow{(5)} \left(\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -1 \\ 0 & 5 & -1 & -5 & 0 \\ \boxed{0} & 0 & 1 & 5 & -10 \\ 0 & 0 & 1 & 5 & -10 \end{array} \right) \xrightarrow{(6)} \\ &\xrightarrow{(6)} \left(\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -1 \\ 0 & 5 & -1 & -5 & 0 \\ 0 & 0 & \textcircled{1} & 5 & -10 \\ \boxed{0} & 0 & 0 & 0 & 0 \end{array} \right) \xrightarrow{(7)} \left(\begin{array}{cccc|c} 1 & 3 & 0 & -2 & -1 \\ 0 & 5 & -1 & -5 & 0 \\ 0 & 0 & 1 & 5 & -10 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right) \end{aligned}$$

Let's see a detailed explanation. We start by defining

$$R_i = i\text{-th row} \quad C_j = j\text{-th column}$$

- 1) We use $m_{11} = 1 \neq 0$ as the "pivot position" and C_1 as the "pivot column". All elements from C_1 below the pivot element are set to zero as follows:

- $R_2 = R_2 - 2 \cdot R_1$
- $R_3 = R_3 - 3 \cdot R_1$
- $R_4 = R_4 - 2 \cdot R_1$

- 2) Since $m_{22} = 0$ (our next pivot element which needs to be different than zero) we transpose R_2 and R_4 .

- 3) We change the sign of R_2 by multiplying the whole row for (-1) .
- 4) The new pivot position is the element $m_{22} = 5 \neq 0$ and the pivot column R_2 . All elements from R_3 below the pivot position are set to zero as follows:
 - $R_3 = R_3 + 2 \cdot R_2$.
- 5) We change the sign of R_3 by multiplying the whole row for (-1) .
- 6) We take $m_{33} = 1 \neq 0$ and C_3 as the new pivot element and column, respectively. All elements in C_3 below the pivot element are set to zero as follows:
 - $R_4 = R_4 - R_3$.
- 7) We remove the last equation R_4 .

The resulting matrix contains three equations (row) and four variables (columns). Thus, the SLE has one degree of freedom or one free variable (λ) that can take any value.

By backward substitution in all equations, we obtain the following general solution of the system that depends on the arbitrary parameter λ :

$$x = -3y + 2\lambda - 1 = 2\lambda + 5$$

$$y = \frac{1}{5}(5\lambda + z) = \frac{1}{5}(5\lambda - 10 - 5\lambda) = -2$$

$$z = -10 - 5\lambda$$

$$t = \lambda$$

The Gaussian elimination algorithm intrinsically provides the rank of a matrix (i.e., number of linearly independent rows) and it coincides with the number of non-zero rows from the echelon matrix $\mathbf{E}$. In the previous example, $\text{rank}(\mathbf{M}) = 3$ since the number of non-zero rows in $\mathbf{E}$ is 3.

Example 8. Using Gaussian elimination to solve a linear system

Consider the following SLE in matrix notation:

$$\begin{pmatrix} 1 & 2 & 1 & 2 \\ -1 & -2 & 3 & 2 \\ 2 & 4 & -1 & 1 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \\ -3/4 \end{pmatrix}$$

The augmented matrix of the SLE is $\mathbf{M} = \left(\begin{array}{cccc|c} 1 & 2 & 1 & 2 & 0 \\ -1 & -2 & 3 & 2 & 1 \\ 2 & 4 & -1 & 1 & -3/4 \end{array} \right)$ Now, let's transform the matrix $\mathbf{M}$ into a triangular matrix. Recall that the Gaussian elimination algorithm consists of transforming a given matrix representing an SLE into another matrix in echelon form that represents an equivalent SLE.

$$\begin{aligned}
 \mathbf{M} &= \left(\begin{array}{cccc|c} \textcircled{1} & 2 & 1 & 2 & 0 \\ -1 & -2 & 3 & 2 & 1 \\ 2 & 4 & -1 & 1 & -\frac{3}{4} \end{array} \right) \approx \left(\begin{array}{cccc|c} 1 & 2 & 1 & 2 & 0 \\ \textcircled{0} & 0 & 4 & 4 & 1 \\ 2 & 4 & -1 & 1 & -\frac{3}{4} \end{array} \right) \approx \left(\begin{array}{cccc|c} 1 & 2 & 1 & 2 & 0 \\ 0 & 0 & 4 & 4 & 1 \\ \textcircled{0} & 0 & -3 & -3 & -\frac{3}{4} \end{array} \right) \approx \\
 &\approx \left(\begin{array}{cccc|c} 1 & 2 & 1 & 2 & 0 \\ 0 & 0 & \textcircled{1} & 1 & \frac{1}{4} \\ 0 & 0 & -3 & -3 & -\frac{3}{4} \end{array} \right) \approx \left(\begin{array}{cccc|c} 1 & 2 & 1 & 2 & 0 \\ 0 & 0 & 1 & 1 & \frac{1}{4} \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)
 \end{aligned}$$

From the last matrix, we could infer:

- a) The last row contains only zeros, so it can be removed.
- b) From the second row, we have $z + t = \frac{1}{4}$, i.e., $z = \frac{1}{4} - t$.
- c) Finally, from the first equation we see that $x + 2y + \left(\frac{1}{4} - t\right) + 2t = 0$, i.e., $x = -t - 2y - \frac{1}{4}$.

As a conclusion: the system is consistent and dependent with 2 degrees of freedom (i.e., two free parameters in the solution). The solution is:

$$\left\{ \begin{array}{lcl} x & = & -t - 2y - \frac{1}{4} \\ y & = & y \\ z & = & \frac{1}{4} - t \\ t & = & t \end{array} \right.$$

7. Cramer's rule for linear systems

Cramer's rule is used to solve SLE whose coefficient matrix, A , is square and its inverse exists. In other words, the SLE needs to have the same number of equations and variables and A a non-zero determinant.

When the coefficient matrix, A , is squared with a non-zero determinant, the associated system, the so-called **Cramer's system** proves that:

- a) It is consistent and independent.
- b) The solution set can be obtained using the following expressions:

$$\forall i = 1, 2, \dots, n \quad x_i = \frac{\det(C_1, \dots, C_{i-1}, B, C_{i+1}, \dots, C_n)}{\det(A)}. \quad (7)$$

In the previous expression, $\det(A)$ represents the determinant of the coefficient matrix while $\det(C_1, \dots, C_{i-1}, B, C_{i+1}, \dots, C_n)$ is the determinant of a matrix resulting from the substitution of the i -th column of A , C_i , with the constant vector, B .

Example 9. Using Cramer's rule to solve linear systems

Let's solve the following SLE (note the equal number of variables and equations):

$$\begin{cases} 3x + y + 2z = 10 \\ 4x + 3y + 4z = 21 \\ 2x + y + 2z = 9 \end{cases}$$

For the system to be solvable using the Cramer's rule, we need to check that the determinant of the coefficient matrix is zero.

$$|A| = \begin{vmatrix} 3 & 1 & 2 \\ 4 & 3 & 4 \\ 2 & 1 & 2 \end{vmatrix} = 18 + 8 + 8 - 12 - 12 - 8 = 2 \neq 0.$$

Now that we know that the SLE is a Cramer's system, we could find the solution set of the system using Cramer's rule:

Constant vector (B) of the system

$$x = \frac{\begin{vmatrix} 10 & 1 & 2 \\ 21 & 3 & 4 \\ 9 & 1 & 2 \end{vmatrix}}{2} = \frac{60 + 36 + 42 - 54 - 40 - 42}{2} = \frac{2}{2} = 1$$

$$y = \frac{\begin{vmatrix} 3 & 10 & 2 \\ 4 & 21 & 4 \\ 2 & 9 & 2 \end{vmatrix}}{2} = \frac{126 + 80 + 72 - 84 - 108 - 80}{2} = \frac{6}{2} = 3$$

$$z = \frac{\begin{vmatrix} 3 & 1 & 10 \\ 4 & 3 & 21 \\ 2 & 1 & 9 \end{vmatrix}}{2} = \frac{81 + 42 + 40 - 60 - 63 - 36}{2} = \frac{4}{2} = 2$$

Hence, the solution is:

$$x = 1 \quad y = 3 \quad z = 2.$$

Observation

If the system is consistent and dependent with rank r smaller than the number of unknown variables n (i.e., the system has $n - r$ degrees of freedom), we could still apply Cramer's rule on a non-zero $r \times r$ minor. From the initial matrix, we could remove all rows and columns unaffected by the minor. The variables associated with the removed columns are considered as parameters (or free variables). Then, for any (fixed) value of the parameters, the SLE is a Cramer's system and has a single unique solution.

Example 10. Using Cramer's rule to solve linear systems

Given the following SLE

$$\left. \begin{array}{l} x + y + z = 1 \\ x - y + z = 2 \\ x + y - z = 3 \\ 3x + y + z = 6 \end{array} \right\}, \text{ the determinant of the augmented matrix is } \begin{vmatrix} 1 & 1 & 1 & 1 \\ 1 & -1 & 1 & 2 \\ 1 & 1 & -1 & 3 \\ 3 & 1 & 1 & 6 \end{vmatrix} = 0$$

and so, its rank is less than 4 being one of its equations a linear combination of the others. On the other hand, the 3rd order minor

$$\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 4$$

and thus, the rank of the augmented matrix and the coefficients matrix is the same (3), yielding a system that is consistent and dependent.

We could remove the last equation (which is a linear combination of the other three) to define a Cramer's system

$$\left. \begin{array}{l} x + y + z = 1 \\ x - y + z = 2 \\ x + y - z = 3 \end{array} \right\} \text{ which can be solved using Cramer's rule as follows:}$$

$$x = \frac{\begin{vmatrix} 1 & 1 & 1 \\ 2 & -1 & 1 \\ 3 & 1 & -1 \end{vmatrix}}{4} = \frac{5}{2}, \quad y = \frac{\begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 3 & -1 \end{vmatrix}}{4} = \frac{-1}{2}, \quad z = \frac{\begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 2 \\ 1 & 1 & 3 \end{vmatrix}}{4} = -1.$$

Example 11. Using Cramer's rule to solve linear systems

Consider the following system

$$\left. \begin{array}{l} x + y + z + t = 1 \\ x - y + z + t = 2 \\ x + y - z + t = 3 \end{array} \right\}, \text{ which has a non-zero 3x3 minor } \begin{vmatrix} 1 & 1 & 1 \\ 1 & -1 & 1 \\ 1 & 1 & -1 \end{vmatrix} = 4$$

and thus its coefficient and augmented matrices have rank 3. Nonetheless, the SLE has 4 variables and so, the system is consistent and dependent with 1 degree of freedom. We could consider the variable t as a parameter (it corresponds to the unused column when computing the above 3x3 minor) and define the following Cramer's system for each value of t

$$\left. \begin{array}{l} x + y + z = 1 - t \\ x - y + z = 2 - t \\ x + y - z = 3 - t \end{array} \right\}$$

The SLE could be solved using:

1) Gaussian elimination:

$$\left(\begin{array}{cccc} 1 & 1 & 1 & 1-t \\ 1 & -1 & 1 & 2-t \\ 1 & 1 & -1 & 3-t \end{array} \right) \rightarrow \left(\begin{array}{cccc} 1 & 1 & 1 & 1-t \\ 1 & -2 & 0 & 1 \\ 0 & 0 & -2 & 2 \end{array} \right) \text{ and we get } x = \frac{5-2t}{2}, y = \frac{-1}{2}, z = -1, t = t$$

2) Cramer's rule:

$$x = \frac{\begin{vmatrix} 1-t & 1 & 1 \\ 2-t & -1 & 1 \\ 3-t & 1 & -1 \end{vmatrix}}{4} = \frac{5-2t}{2}, \quad y = \frac{\begin{vmatrix} 1 & 1-t & 1 \\ 1 & 2-t & 1 \\ 1 & 3-t & -1 \end{vmatrix}}{4} = \frac{-1}{2}, \quad z = \frac{\begin{vmatrix} 1 & 1 & 1-t \\ 1 & -1 & 2-t \\ 1 & 1 & 3-t \end{vmatrix}}{4} = -1, \quad t = t$$

8. Geometric interpretation of SLE

Before diving into the three dimensional space (SLE with three variables), it is useful to briefly introduce the two dimensional plane (SLE with two variables). In a system of m linear equations with 2 unknown variables (x and y), each equation represents a line in the plane such that:

- If the system is consistent and independent, there only exists a single value (x_0, y_0) as a solution of the system (and it's unique). Hence, (x_0, y_0) simultaneously belongs to all the m lines which means that all lines intersect at this point.
- If the system is consistent and dependent, all lines share infinitely many points, meaning they are coincident (they are actually the same line expressed in different ways).
- If the system is inconsistent, there is no shared point between the m lines. In the specific case of $m = 2$, both lines will be parallel and non-coincident.

In the 3D space, a plane π is determined by a linear equation with three variables (x, y, z) . Considering $a, b, c, d \in \mathbb{R}$ as scalar values, the general expression is written as:

$$\pi : a \cdot x + b \cdot y + c \cdot z = d \quad (8)$$

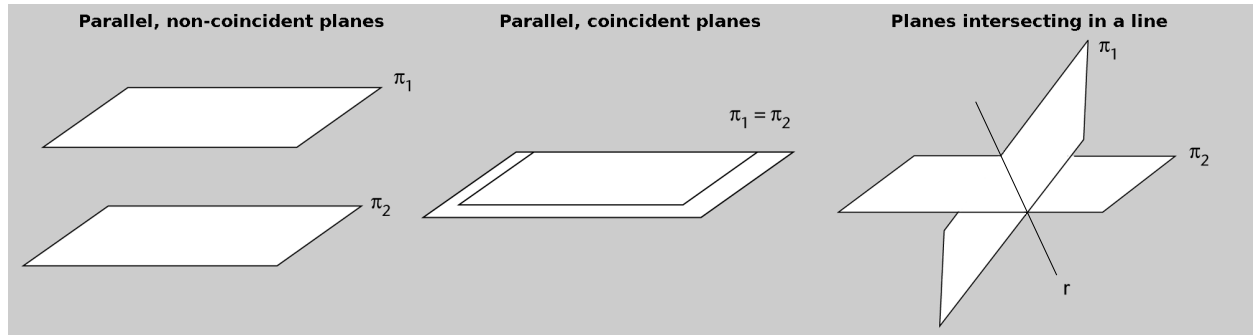
A SLE with m equations and 3 variables could be interpreted as a set of m planes in a 3D space. In this case, the Rouché-Capelli theorem is used to study the relative position of the m planes (i.e., whether they intersect at a point or a line, if they are parallel, etc.).

If the SLE is consistent and independent, there exists only one solution (x_0, y_0, z_0) that simultaneously satisfies all m equations. In other words, (x_0, y_0, z_0) is the point of intersection between all planes, as it is the unique point in the space that belongs to all planes at the same time. We illustrate this situation in Example 12 and discuss other possible cases in the subsequent examples.

Example 12. Intersection of two planes

Consider the case of $m = 2$ (two planes). Note that the only possible relative positions between the planes are shown in figure 3: they are either parallel and non-coincident (they do not intersect), parallel and coincident (they share infinitely many points thus they are the same plane) or they intersect at an infinite number of points that draw a line r . Hence, with $m = 2$, they can't intersect at a single point (something we already knew in advance because the system has, at least, one degree of freedom).

Figure 3. All different relative positions between two planes in a 3D space.

**Example 13. Intersection of three planes in a point**

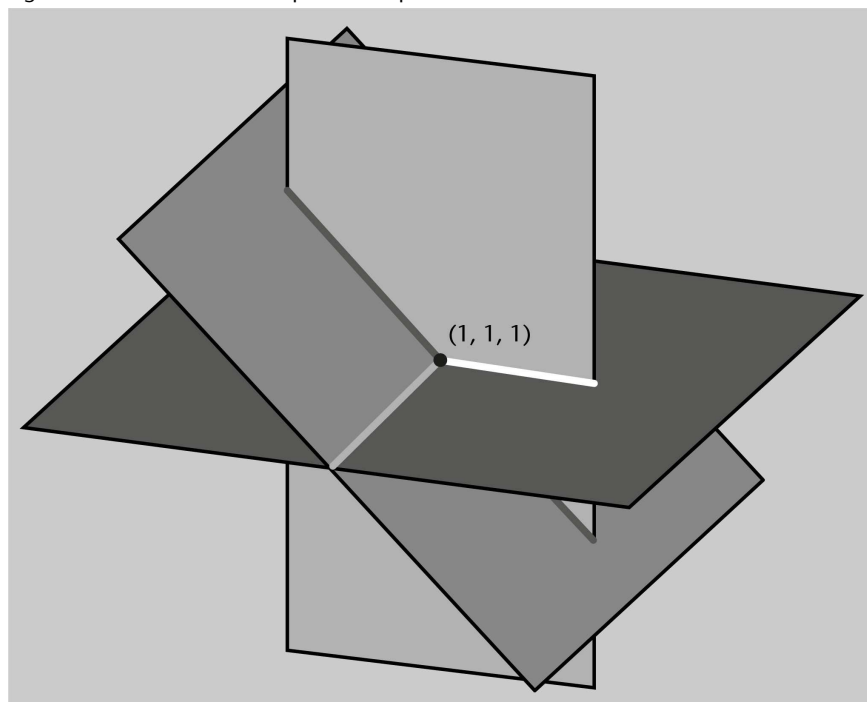
Consider the following system of linear equations:

$$\begin{cases} x + y = 2 \\ y + z = 2 \\ x + z = 2 \end{cases}$$

Easily, we infer that the system is consistent and independent ($\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 3$). The only solution of this system is $(x, y, z) = (1, 1, 1)$

Figure 4 shows that the solution of an independent system could be seen as the intersection of three planes defined by the three linear equations of the system.

Figure 4. Intersection of three planes in a point.



Example 14. Intersection of two planes in a line

Consider the following system of linear equations:

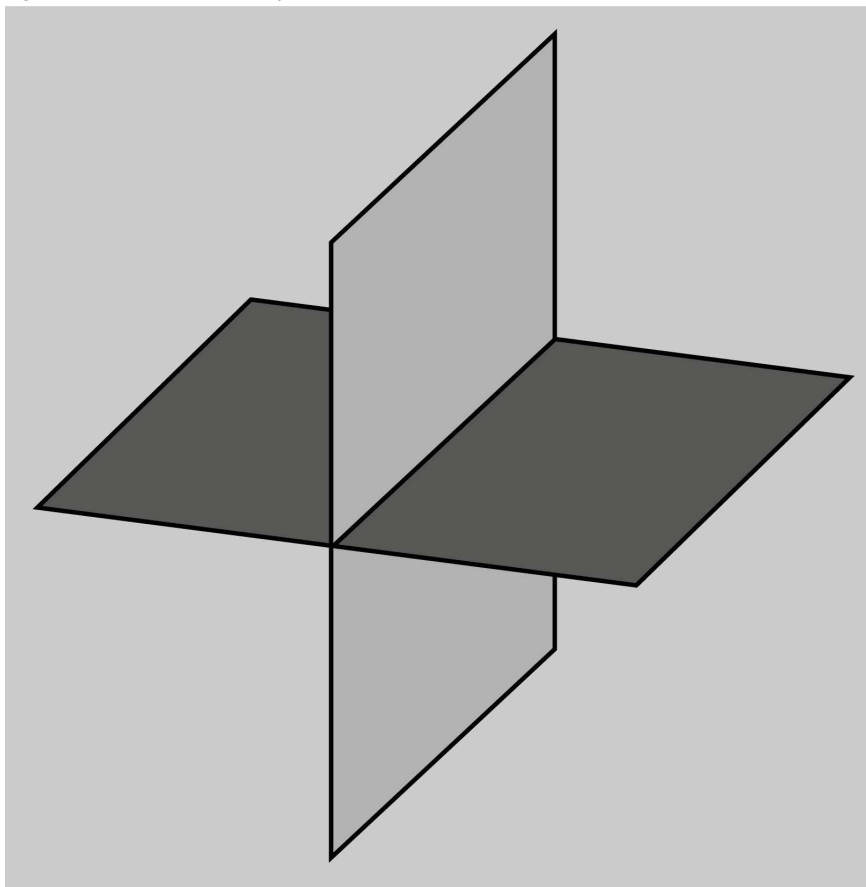
$$\begin{cases} x - y + z = 0 \\ 2x + y - z = 3 \end{cases}$$

In this example, there are three variables but only two linearly independent equations. Hence, the system is consistent and dependent as $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 2 < 3$.

Figure 5 shows the associated plane to each equation and the intersection of both (i.e., all points that jointly satisfy both equations) which represents a line.

The parameterised solution of this system is $(x, y, z) = (1, z + 1, z)$

Figure 5. Intersection of two planes in a line.

**Example 15. Intersection of three planes in a line**

Consider the following SLE:

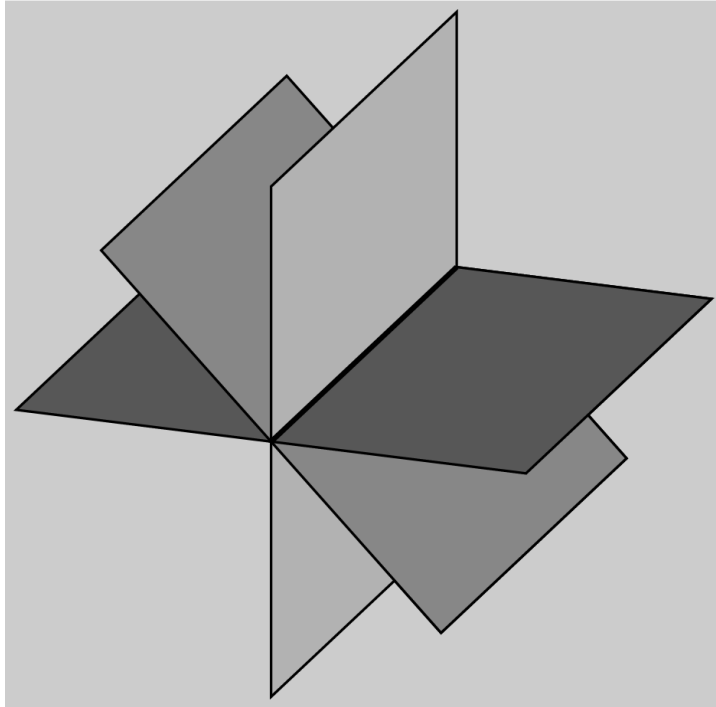
$$\begin{cases} x - y + z = 0 \\ 2x + y - z = 3 \\ 3x = 3 \end{cases}$$

Here, the first two equations are exactly the same as in the previous example, while the third is a linear combination of both (concretely, the summation of the first and second equations). Hence, it is obvious that, again, the system is consistent and dependent ($\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 2 < 3$).

Figure 6 shows the two planes from the previous example representing the first two equations and adds a third plane that intersects at the exact same line.

The parameterised solution of this system is still $(x, y, z) = (1, z + 1, z)$

Figure 6. Intersection of three planes in a line.



Example 16. Two parallel, non-coincident planes

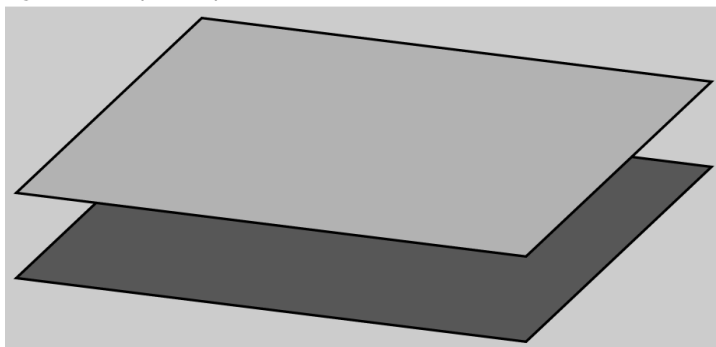
Consider the following linear system:

$$\begin{cases} x - y + z = 0 \\ x - y + z = 1 \end{cases}$$

Quickly, we note that the system is inconsistent since it can't satisfy both equations simultaneously ($x - y + z$ can't be 0 and 1 at the same time). Computing the rank of the matrices we realise that $\text{rank}(\mathbf{A}) = 1$ and $\text{rank}(\mathbf{M}) = 2$. Hence the system has no solution.

Geometrically, it indicates that both planes (representing both equations) never intersect. In other words, both planes are parallel and non-coincident, as depicted in figure 7.

Figure 7. Two parallel planes.



Example 17. Three non-intersecting planes

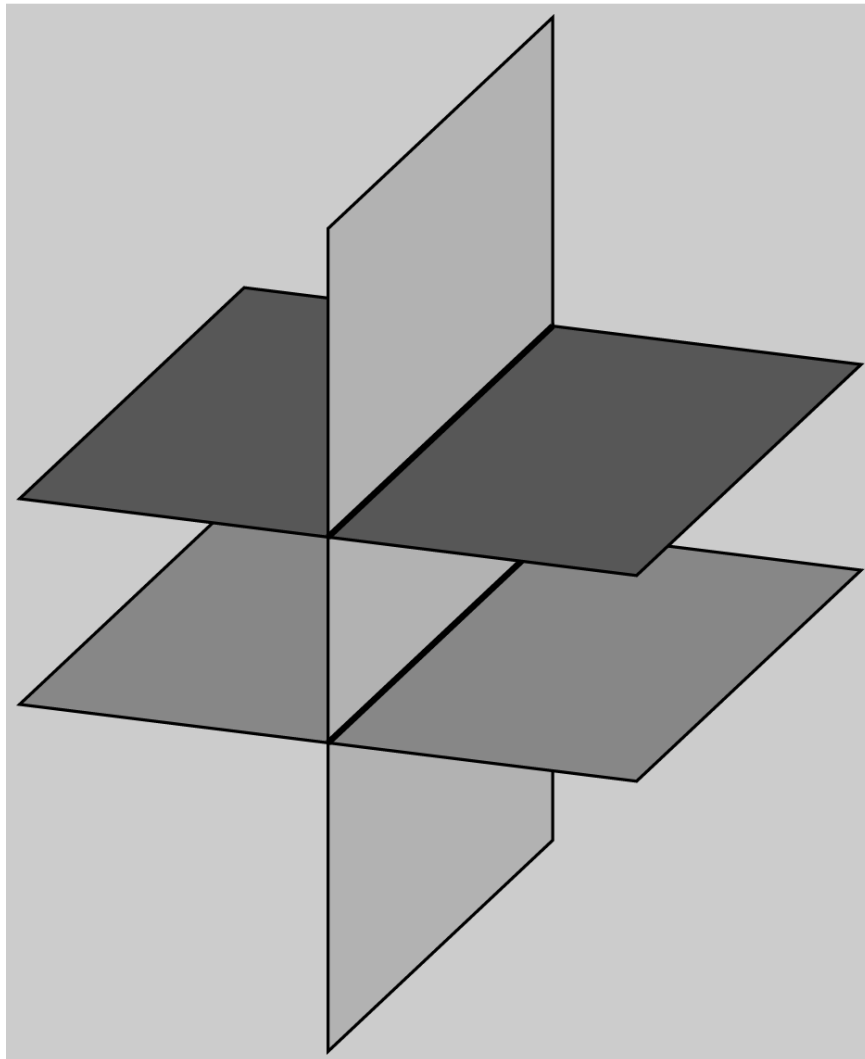
Consider the following SLE:

$$\begin{cases} x - y + z = 0 \\ 2x + y - z = 3 \\ 2x + y - z = 0 \end{cases}$$

If we considered only the first two equations or the first and last equations, we would have a consistent and independent system, representing two planes that intersect at a line. However, since the planes associated with the second and third equations are parallel, a system using both equations is inconsistent. Hence, there isn't a solution set that satisfies the second and third equation regardless of the first, so the system of 3 equations considered here is also inconsistent.

Indeed, if we compute the rank of the associated matrices we see that $\text{rank}(\mathbf{A}) = 2 < \text{rank}(\mathbf{M}) = 3$. The geometric representation of the previous explanation is shown in figure 8.

Figure 8. Three planes without a common point.



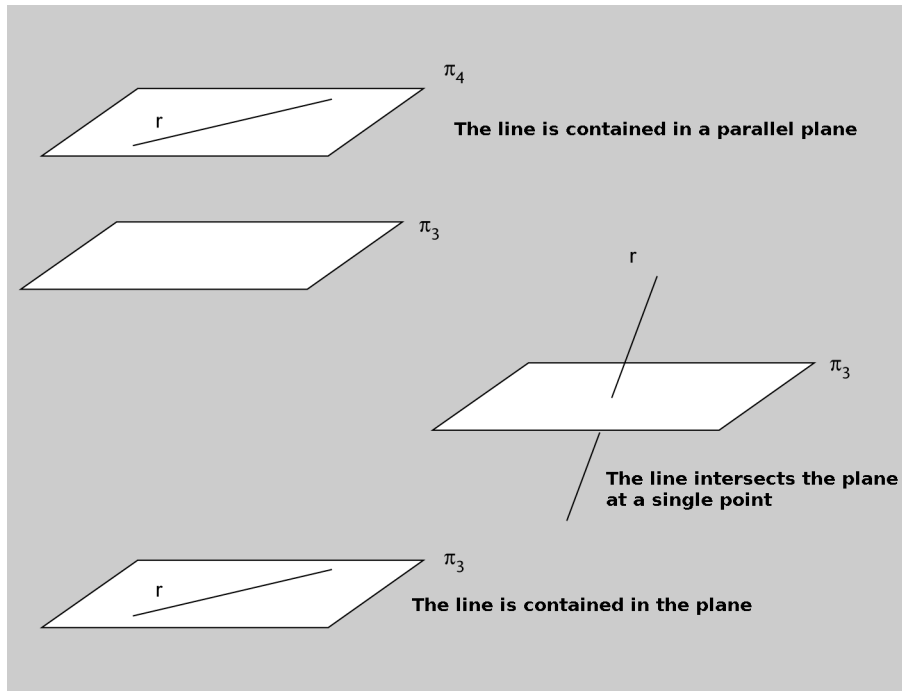
In some of the previous examples, we have shown that the intersection between two non-coincident planes, π_1 and π_2 , is a line in the space r . Thereupon, the equation of a line is often expressed as a consistent and dependent system with 2 equations and 3 variables, i.e.,

$$r : \begin{cases} \pi_1 : & a_1x + b_1y + c_1z = d_1 \\ \pi_2 : & a_2x + b_2y + c_2z = d_2 \end{cases} \quad (9)$$

Consequently, the relative position between a line r (given the equations for π_1 and π_2) and a plane π_3 is studied by discussing the corresponding system of 3 equations with 3 variables:

- When the system is inconsistent (i.e., there is no shared point between the plane and the line), the line r is defined on a plane π_4 parallel (and non-coincident) to π_3 .
- When the system is consistent and independent (i.e., the line and the plane share a single point), the line r intersects with the plane π_3 at a single point.
- When the system is consistent and dependent (i.e., the line and the plane have infinitely many points in common), the line r is defined on the plane π_3 .

Figure 9. Different options for the relative position between a line and a plane in the space.



Similarly, we could also study the relative position between two lines, r_1 and r_2 , in the 3D space: each line is defined by the intersection of two planes,

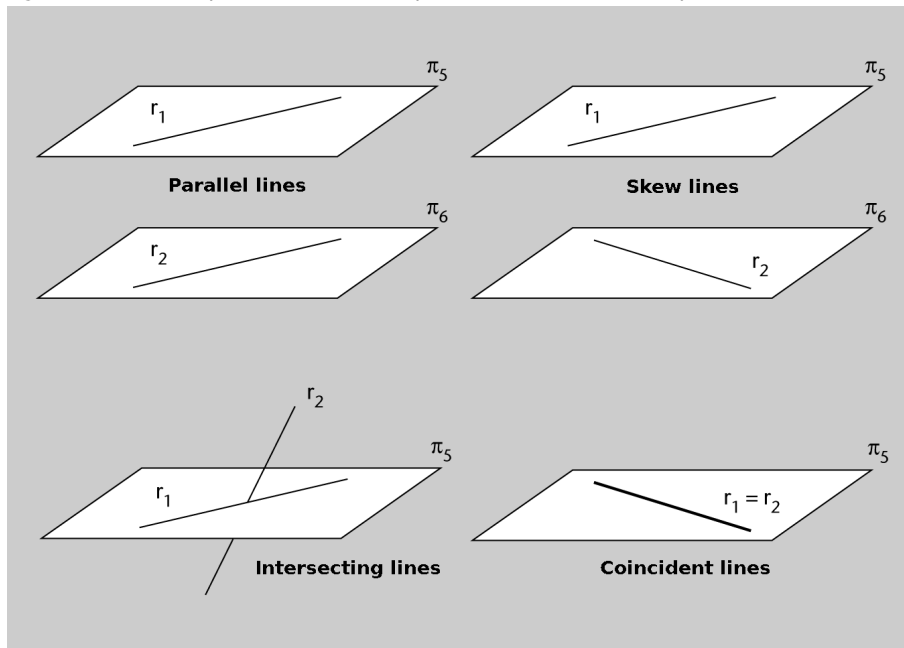
therefore each line is determined by a system of 2 equations with 3 variables (consistent and dependent) constituting a compounded system of 4 equations (with 3 variables).

$$\begin{aligned} r_1 : \begin{cases} \pi_1 : & a_1x + b_1y + c_1z = d_1 \\ \pi_2 : & a_2x + b_2y + c_2z = d_2 \end{cases} \\ r_2 : \begin{cases} \pi_3 : & a_3x + b_3y + c_3z = d_3 \\ \pi_4 : & a_4x + b_4y + c_4z = d_4 \end{cases} \end{aligned} \quad (10)$$

Discussing the compounded system, we argue that:

- When the system is inconsistent (no shared point between the lines), r_1 and r_2 are either parallel (non coincident) or they cross — see figure 10. In both cases, each line belongs to two parallel and non coincident planes π_5 and π_6 .
- When the system is consistent and independent (one shared point between the lines), r_1 intersect r_2 at a single point.
- When the system is consistent and dependent (infinite number of shared points between the lines), r_1 and r_2 are the same line (parallel and coincident).

Figure 10. Different options for the relative position of two lines in the space.



Summary

In this module, we have presented the main ideas and results related to systems of linear equations. The key concepts introduced here are:

- Definition of a system of linear equations (SLE) and its components: coefficients, variables and constants. Equivalent systems.
- Classification of systems with respect the number of solutions: consistent systems (dependent and independent) and inconsistent systems.
- Express a SLE using matrix notation: coefficient and augmented matrices.
- Discussion of a SLE. Rouché-Capelli theorem.
- Homogeneous SLE. Trivial solution.
- Gaussian elimination algorithm.
- Cramer's systems and Cramer's rule.
- Geometric interpretation of a SLE in the plane (2 variables) and in the space (3 variables): study the relative position between lines and planes.

Finally, this module contains several examples and activities with detailed solutions (with and without using mathematical software). Some of them are further illustrated with practical applications in different fields.

Self-assessment exercises

1. Compute the rank of matrix A as a function of parameter s :

$$A = \begin{pmatrix} s+3 & 1 & 2 & -s \\ s & s-1 & 1 & 2s \\ 3s+3 & s & s+3 & 3 \end{pmatrix}$$

With and without the help of mathematical software.

For $s = 0$, find the solution(s) of the linear system

$$A \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 2 \\ 1 \\ 3 \end{pmatrix}$$

2. Discuss the following system with respect all values of parameter $k \in \mathbb{R}$:

$$\begin{cases} kx + y + z = k \\ x + ky + z = k \\ x + y + kz = k \end{cases}$$

3. Find the parameter a and b values such that the following homogeneous system has other solutions than the trivial

$$\begin{cases} x + 2y - 3z = 0 \\ 2x + 5y - 8z = 0 \\ ax + by + 3z = 0 \\ ax + y + bz = 0 \end{cases}$$

4. A well-known company manufactures three different products: A, B and C. The following tables shows a summary of sales and profits of these three along the last 3 years.

Year	Units sold for product A	Units sold for product B	Units sold for product C	Total profit
1	100	500	200	39700
2	300	400	300	45600
3	200	800	500	73300

Formulate the corresponding system of equations and solve it using Gaussian elimination to find the profit per unit of each product A, B and C sold.

5. Discuss the following system as a function of parameters a and b and solve it for those cases that the system is consistent:

$$\begin{cases} 3x - y + 2z = 1 \\ x + 4y + z = b \\ 2x - 5y + az = -2 \end{cases}$$

Observation

The profit for every unit of each product does not change during the 3-year period studied.

6. Find the parameter a values for which the system is consistent and find the solutions (if any)

$$\begin{cases} x - 2y + az = 1 - a \\ x + 4y + a^2z = 6 \\ x - 8y + a^2z = -6 \end{cases}$$

7. Find the relative position of the following two planes:

$$\pi_1 : 2x + 3y - z + 8 = 0 \quad \pi_2 : -4x - 6y + 2z - 16 = 0$$

With and without the help of mathematical software.

8. Find the parameter value m for which the line r is parallel to the plane π :

$$r : \frac{x-1}{2} = \frac{y+5}{4} = \frac{z-3}{2} \quad \pi : 4x + my + z - 2 = 0$$

With and without the help of mathematical software.

9. Find the parameter value a such that the lines r and r' intersect:

$$r : \frac{x-2}{5} = \frac{y}{6} = \frac{z+1}{2} \quad r' : \frac{x-a}{2} = \frac{y+1}{3} = \frac{z-2}{3}$$

With and without the help of mathematical software.

10. Study and solve the following linear system:

$$\begin{cases} x - 2y = 3 \\ 2x - 4y + 2z - 4t = 4 \\ -3x + 6y + z - 2t = -10 \end{cases}$$

With and without the help of mathematical software.

11.

- a) Discuss the following system with respect all values for parameter $k \in \mathbb{R}$

$$\begin{cases} x + y + kz = 1 \\ x + 2y + 8z = k \\ x + y + 7z = 1 \\ 2x + y + kz = 2 \end{cases}$$

- b) Use Gaussian elimination to solve the system in the case(s) for which the system is consistent.
c) Consider two lines, r and s in the space $\mathbb{R}^3$:

$$r : \begin{cases} x + y + kz = 1 \\ x + 2y + 8z = k \end{cases}, \quad s : \begin{cases} x + y + 7z = 1 \\ 2x + y + kz = 2 \end{cases}$$

Use the results from (a) to find the parameter k values for which both lines intersect.

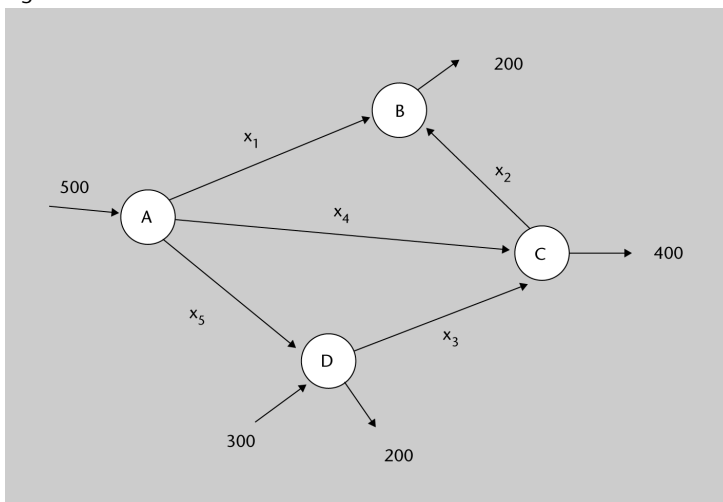
12. Discuss the following system as a function of parameters a and b . When the system is consistent, find the solution set using Cramer's rule

$$\begin{cases} ax + by + z = 1 \\ x + aby + z = b \\ x + by + az = 1 \end{cases}$$

Each equation resembles a plane in the space. What is their relative position with respect the parameter (a,b) values?

13. The following scheme in Figure 11 represents a multi-repeater network for which data flows as indicated by the arrow connections. Historical data collection shows the data flow (in MB/h) send to and from each node.

Figure 11



Formulate, discuss and solve (if possible) the corresponding linear system of equations in the situation of equilibrium (that is, the data rate incoming to and outgoing from each node is the same)

14. The distribution of resources is a common problem in many branches of Science: both for the inventory of resources as well as finding their optimal distribution. Those resources are typically physical (in the sense of measurable with some quantity) and human (for example, a football coach needs to decide the best possible line-up for which the performance of the team is optimal). In this setting, matrix notation helps solving the systems.

Consider a computer scientist supervising the deployment of 4 different types of networks in a company. Each network is associated to a different service that use different resources, as indicated in the following table:

With the help of mathematical software.

Service type	Printers	Scanners	PC	USB sticks
A	1	2	2	0
B	10	1	1	1
C	0	4	1	0
D	10	0	1	1

The company has the following available hardware: 152 printers, 550 scanners, 4 USB sticks and 309 PC. How many services of each type could be used at the same time?

15. A student takes photographs using a digital camera. The size of each photograph depends on its quality: 0.2 MB for normal quality and a MB for optimal quality images. This week, the student wants to print a total of 24 photographs which take up a total of 9.2 MB of memory.

Answer the following items:

- Formulate a system of equations (as a function of parameter a) in which the variables (unknown) are the number of images of each quality type. Discuss the resulting system and use Cramer's rule to find the solution sets when the system is consistent.
- Is there any unfeasible size (in MB) for optimal quality images?
- Find the optimal quality size (in MB) for which the number of normal and optimal quality images is the same.

16. Let's imagine you are the system administrator of a local area network (LAN) and you want to maximise the disk allocation to users of the network under some restrictive conditions. There are two different types of hard drives in the market:

- HD-A has 300 GB of space, costs 200 €, needs 1 hour per week of maintenance work and consumes 15 units of electricity daily.
- HD-B has 200 GB of capacity, costs 100 €, needs 2 hours per week of maintenance work and consumes 3 units of electricity daily.

The total budget for the project is 1500 € and could take up to 100 units of electricity. Moreover, you can only dedicate 27 hours in maintenance work weekly.

How many hard drives of each type are you going to buy to maximise the amount of memory available in the LAN? Formulate the problem using linear equations and inequalities.

17. Consider the following two matrices:

$$A(k) = \begin{pmatrix} 6 & 0 & 0 & 0 & 0 & 0 & 0 \\ 9 & 12k-15 & -12k+3 & 15 & 12k-3 & 0 & -9 \\ 2k-13 & 2k-5 & 2k+9 & -1 & 4k+3 & 0 & -13 \\ 30 & 12k-18 & -12k+18 & 18 & 12k-18 & 0 & 6 \\ 2k-16 & -4k+16 & 8k & -16 & -2k+12 & 0 & -4 \\ 22k-5 & -14k+11 & 16k-3 & 6k-5 & -16k+3 & 6k+12 & -5 \\ 0 & -6k+6 & 6k-6 & 0 & -6k+6 & 0 & 12 \end{pmatrix}, \quad B = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ -4 \\ 0 \end{pmatrix}.$$

- Compute the determinant of matrix A .
- Indicate for which values of parameter k the matrix A does not have maximum rank.
- Determine the rank of matrix A as a function of parameter $k \in \mathbb{R}$ values.
- Discuss the system $AX = B$ as a function of parameter value $k \in \mathbb{R}$, being X a column vector containing variables $x_1, x_2, \dots, x_7$.

18. Consider the following planes in the space:

$$\pi_1 : x + y + z = 2 \quad \pi_2 : 2x + 3y + z = 3 \quad \pi_3 : kx + 10y + 4z = 11$$

- Find the parameter k value(s) for which the intersection of the three planes defines a line in the space. For the same k values, solve the resulting SLE using Gaussian elimination (check the result with mathematical software).

Note

Formulate the system of equations and use Gaussian elimination algorithm to find the solution.

With the help of mathematical software.

- b)** Find the parameter k value for which the intersection of the three planes defines a point in the space. For $k = 10$, determine the coordinates of the point using Cramer's rule (check the result with mathematical software).
- 19.** Consider the following system of equations:

$$\begin{cases} 3x + ky + kz &= 0 \\ x - y &= 0 \\ 3x - 2y &= 0 \end{cases}$$

- a)** Discuss the SLE for all values of parameter $k \in \mathbb{R}$.
- b)** For $k = 1$, solve the system using Gaussian elimination.
- 20.** Given the following system of equations (three planes in $\mathbb{R}^3$):

$$\begin{cases} x - 3y + z &= 0 \\ 3x + 6y + z &= 3 \\ 2x - y + z &= 1 \end{cases}$$

- a)** Try solving the system using Cramer's rule.
- b)** What is the relative position of the three planes (intersection at a point, intersection at a line, parallel, etc.)?

Solutions

Solution to the initial example

1.

- a) Using the equilibrium condition that the incoming and outgoing data flow in each node is identical, we build the following SLE:

$$\left\{ \begin{array}{l} 600 = x + z \\ x = y + t \\ y + u = 500 \\ z + v = 600 \\ t + w = v \\ 500 = w + u \end{array} \right. \quad \text{or, equivalently} \quad \left\{ \begin{array}{l} x + z = 600 \\ x - y - t = 0 \\ y + u = 500 \\ v + z = 600 \\ t - v + w = 0 \\ u + w = 500 \end{array} \right.$$

This SLE has 7 variables (corresponding to each connection) and 6 equations (corresponding to each node). Studying the rank of the system we conclude that the SLE is consistent and dependent, with two degrees of freedom. Hence, it has infinitely many solutions. Using (y, z) as parameters, the solution is:

$$(x, y, z, t, u, v, w) = (600 - z, y, z, 600 - y - z, 500 - y, 600 - z, y).$$

Any mathematical software can help finding the solution.

- b) Now, the new system has 5 variables

$$\left\{ \begin{array}{l} x + z = 600 \\ x - y - t = 0 \\ y = 200 \\ z = 500 \\ t + w = 100 \\ w = 200 \end{array} \right. ,$$

and is consistent and independent. The solution is

$$(x, y, z, t, w) = (100, 200, 500, -100, 200).$$

Note that, in this case $t = -100$. This indicates that the data flow direction is reversed: instead of going from node B to E (like in the schematic), it actually goes from E to B.

Solution to self-assessment exercises

1. To compute the rank of matrix **A**, we initially compute the value of the 3x3 minor composed of the first three columns:

$$\begin{vmatrix} s+3 & 1 & 2 \\ s & s-1 & 1 \\ 3s+3 & s & s+3 \end{vmatrix} = (s+3)^2(s-1) + (3s+3) + 2s^2 - 6(s+1)(s-1) - 2s^2 - 6s$$

$$= (s-1) \left[(s+3)^2 - 3 - 6(s+1) \right] = s^2(s-1)$$

The roots of the resulting polynomial are $\{0,1\}$. Hence, for all other parameter values ($s = \mathbb{R} \setminus \{0,1\}$) the matrix rank is 3. For $s = 0,1$ there may be other non-zero 3×3 minor. Let us study the rank using Gaussian elimination:

- Case $s=0$: operating the matrix, we get a rank of 3

$$\begin{pmatrix} 3 & 1 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 3 & 0 & 3 & 3 \end{pmatrix} \xrightarrow{(1)} \begin{pmatrix} 3 & 1 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & -1 & 1 & 3 \end{pmatrix} \xrightarrow{(2)} \begin{pmatrix} 3 & 1 & 2 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{pmatrix}$$

(1) we subtract first row from the third row ($R_3 = R_3 - R_1$)

(2) we subtract second row from the third row ($R_3 = R_3 - R_2$)

- Case $s=1$: operating the matrix, we get a rank of 2.

$$\begin{pmatrix} 4 & 1 & 2 & -1 \\ 1 & 0 & 1 & 2 \\ 6 & 1 & 4 & 3 \end{pmatrix} \xrightarrow{(1)} \begin{pmatrix} 1 & 0 & 1 & 2 \\ 4 & 1 & 2 & -1 \\ 6 & 1 & 4 & 3 \end{pmatrix} \xrightarrow{(2)} \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & -9 \\ 0 & 1 & -2 & -9 \end{pmatrix} \xrightarrow{(3)} \begin{pmatrix} 1 & 0 & 1 & 2 \\ 0 & 1 & -2 & -9 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

(1) we permute the first and second rows.

(2) we subtract four and six times the first row to the second and third rows, respectively.

$$R_2 = R_2 - 4R_1 \quad \text{and} \quad R_3 = R_3 - 6R_1$$

(3) we subtract the second row from the third row ($R_3 = R_3 - R_2$)

$$\text{To sum up, the rank}(\mathbf{A}) = \begin{cases} 3 & \text{if } s \neq 1 \\ 2 & \text{if } s = 1 \end{cases}$$

Now, we need to solve the system for $s = 0$. Since we already know that the rank of $\mathbf{A}$ is 3, the SLE is compatible and dependent, with one degree of freedom. To find the solutions, we use Gaussian elimination:

$$\left(\begin{array}{cccc|c} 3 & 1 & 2 & 0 & 2 \\ 0 & -1 & 1 & 0 & 1 \\ 3 & 0 & 3 & 3 & 3 \end{array} \right) \xrightarrow{(1)} \left(\begin{array}{cccc|c} 3 & 1 & 2 & 0 & 2 \\ 0 & -1 & 1 & 0 & 1 \\ 0 & -1 & 1 & 3 & 1 \end{array} \right) \xrightarrow{(2)} \left(\begin{array}{cccc|c} 3 & 1 & 2 & 0 & 2 \\ 0 & -1 & 1 & 0 & 1 \\ 0 & 0 & 0 & 3 & 0 \end{array} \right)$$

(1) we subtract first row from the third row ($R_3 = R_3 - R_1$).

(2) we subtract second row from the third row ($R_3 = R_3 - R_2$).

From the third row, we know that $t = 0$, and from the other two $x = 1 - z$ and $y = z - 1$, being $z = z$.

2. First and foremost, we derive the coefficient and augmented matrices:

$$\mathbf{A} = \begin{pmatrix} k & 1 & 1 \\ 1 & k & 1 \\ 1 & 1 & k \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{ccc|c} k & 1 & 1 & k \\ 1 & k & 1 & k \\ 1 & 1 & k & k \end{array} \right)$$

This system has 3 equations and 3 variables and thus, the rank of the matrices is at most 3. To have maximum rank, the determinant of the coefficient matrix needs to be non-zero:

$$|\mathbf{A}| = (k^3 + 1 + 1) - (k + k + k) = k^3 - 3k + 2 = (k - 1)^2(k + 2)$$

The determinant is a polynomial function of parameter k . The roots of the polynomial make the determinant zero: $|\mathbf{A}| = 0 \iff k = \{1, -2\}$. Hence, for any other value ($k \in \mathbb{R} \setminus \{1, -2\}$), the system has maximum rank and is consistent and independent ($\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = n = 3$).

Let us study the other two cases separately:

- Case $k = -2$: The matrices are:

$$\mathbf{A} = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{ccc|c} -2 & 1 & 1 & -2 \\ 1 & -2 & 1 & -2 \\ 1 & 1 & -2 & -2 \end{array} \right)$$

We already know that $\text{rank}(\mathbf{A}) < 3$ because $|\mathbf{A}| = 0$ but we need to compute the $\text{rank}(\mathbf{M})$. For that, we need to check 3x3 minors of the augmented matrix. For example, using the last three columns we get a non-zero minor:

$$\begin{vmatrix} 1 & 1 & -2 \\ -2 & 1 & -2 \\ 1 & -2 & -2 \end{vmatrix} = (-2 - 8 - 2) - (-2 + 4 + 4) = -18 \neq 0$$

Consequently, $\text{rank}(\mathbf{M}) = 3 > \text{rank}(\mathbf{A})$ and thus, the system is inconsistent.

- Case $k = 1$: The matrices are:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 \end{array} \right)$$

We clearly see that all rows are identical (each row is a trivial linear combination of the others). Since there are 1x1 minors different than zero (i.e. the matrix is not all zeros), the rank of $\mathbf{A}$ and $\mathbf{M}$ is 1. Hence, the system is consistent and dependent with 2 degrees of freedom ($n - \text{rank}(\mathbf{M}) = 3 - 1 = 2$).

To sum up, the system is $\begin{cases} \text{consistent and independent} & \text{if } k \neq 1, -2 \\ \text{consistent and dependent} & \text{if } k = 1 \\ \text{inconsistent} & \text{if } k = -2 \end{cases}$

3. The coefficient matrix of the system is:

$$\mathbf{A} = \begin{pmatrix} 1 & 2 & -3 \\ 2 & 5 & -8 \\ a & b & 3 \\ a & 1 & b \end{pmatrix}$$

The system is homogeneous and has at least one solution, the trivial $(x,y,z) = (0,0,0)$. However, we are interested in finding parameter values (a,b) such that the system contains also nontrivial solutions. Mathematically, we need to find a and b such that $\text{rank}(\mathbf{A}) < 3$ (otherwise the system will have only one solution, the trivial), that is, force all 3×3 minors to be zero.

Carefully looking at the matrix, we see that there already is a non-zero 2×2 minor $\begin{vmatrix} 1 & 2 \\ 2 & 5 \end{vmatrix} = 1 \neq 0$, so the system has at least rank 2. We extend this minor to form two 3×3 minors and build a system of two equations (that's because we have two unknown parameters, a and b).

$$\begin{vmatrix} 1 & 2 & -3 \\ 2 & 5 & -8 \\ a & b & 3 \end{vmatrix} = (15 - 6b - 16a) - (-15a + 12 - 8b) = -a + 2b + 3 = 0$$

$$\begin{vmatrix} 1 & 2 & -3 \\ 2 & 5 & -8 \\ a & 1 & b \end{vmatrix} = (5b - 6 - 16a) - (-15a + 4b - 8) = -a + b + 2 = 0$$

$$\begin{cases} -a + 2b = -3 \\ -a + b = -2 \end{cases}$$

The solution of this system is: $b = -1, a = 1$

Finally, computing all the other 3×3 minors from the coefficient matrix, we see that they are all zero. Hence, the initial SLE has nontrivial solutions for $b = -1$ and $a = 1$

4. The variables of this system are:

- x , the company's profit for each unit of product A sold.
- y , the company's profit for each unit of product B sold.
- z , the company's profit for each unit of product C sold.

Then the system could be defined using these three variables and three equations as follows:

$$\begin{cases} 100x + 500y + 200z = 39700 \\ 300x + 400y + 300z = 45600 \\ 200x + 800y + 500z = 73300 \end{cases}$$

We use Gaussian elimination to solve the system:

$$\left(\begin{array}{ccc|c} 100 & 500 & 200 & 39700 \\ 300 & 400 & 300 & 45600 \\ 200 & 800 & 500 & 73300 \end{array} \right) \xrightarrow{(1)} \left(\begin{array}{ccc|c} 1 & 5 & 2 & 397 \\ 3 & 4 & 3 & 456 \\ 2 & 8 & 5 & 733 \end{array} \right) \xrightarrow{(2)}$$

$$\left(\begin{array}{ccc|c} 1 & 5 & 2 & 397 \\ 0 & -11 & -3 & -735 \\ 0 & -2 & 1 & -61 \end{array} \right) \xrightarrow{(3)} \left(\begin{array}{ccc|c} 1 & 5 & 2 & 397 \\ 0 & -11 & -3 & -735 \\ 0 & 0 & 17 & 799 \end{array} \right)$$

(1) we divide each row by 100

(2) $R_2 = R_2 - 3R_1$ and $R_3 = R_3 - 2R_1$

(3) $R_3 = 11R_3 - 2R_2$

The corresponding system after reducing the matrix in echelon form is:

$$\begin{cases} x + 5y + 2z = 397 \\ -11y - 3z = -735 \\ 17z = 799 \end{cases} \Rightarrow z = 799/17 = 47$$

$$-11y - 3 \cdot 47 = -735 \Rightarrow y = \frac{-735 + 3 \cdot 47}{-11} = 54$$

$$x + 5 \cdot 54 + 2 \cdot 47 = 397 \Rightarrow x = 397 - 5 \cdot 54 - 2 \cdot 47 = 33$$

Hence, the company's profit for each unit sold is:

$x = 33$ euros for each unit of product A sold.

$y = 54$ euros for each unit of product B sold.

$z = 47$ euros for each unit of product C sold.

5. The coefficient and augmented matrices are:

$$\mathbf{A} = \begin{pmatrix} 3 & -1 & 2 \\ 1 & 4 & 1 \\ 2 & -5 & a \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{ccc|c} 3 & -1 & 2 & 1 \\ 1 & 4 & 1 & b \\ 2 & -5 & a & -2 \end{array} \right)$$

The rank of the coefficient matrix as a function of the parameter a is (note that parameter b is only in the constant vector):

$$|\mathbf{A}| = \begin{vmatrix} 3 & -1 & 2 \\ 1 & 4 & 1 \\ 2 & -5 & a \end{vmatrix} = 13a - 13. \quad |\mathbf{A}| = 0 \iff a = 1$$

Hence, we need to discuss two different cases:

- Case $a \neq 1$. In this case, the rank of the system is maximum and equal to the number of variables: $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 3 = n$. Therefore, the system is consistent and independent (single solution).

We could use Cramer's rule to solve the system:

$$x = \frac{\begin{vmatrix} 1 & -1 & 2 \\ b & 4 & 1 \\ -2 & -5 & a \end{vmatrix}}{13a - 13} = \frac{4a - 10b + ab + 23}{13a - 13}$$

$$y = \frac{\begin{vmatrix} 3 & 1 & 2 \\ 1 & b & 1 \\ 2 & -2 & a \end{vmatrix}}{13a - 13} = \frac{-a - 4b + 3ab + 4}{13a - 13}$$

$$z = \frac{\begin{vmatrix} 3 & -1 & 1 \\ 1 & 4 & b \\ 2 & -5 & -2 \end{vmatrix}}{13a - 13} = \frac{13b - 39}{13a - 13}$$

- Case $a = 1$. Now, the coefficient and augmented matrices are:

$$\mathbf{A} = \begin{pmatrix} 3 & -1 & 2 \\ 1 & 4 & 1 \\ 2 & -5 & 1 \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{ccc|c} 3 & -1 & 2 & 1 \\ 1 & 4 & 1 & b \\ 2 & -5 & 1 & -2 \end{array} \right)$$

Firstly, we study the rank of $\mathbf{A}$, which we know is smaller than 3:

$$\left. \begin{array}{l} |\mathbf{A}| = 0 \\ \begin{vmatrix} 3 & -1 \\ 1 & 4 \end{vmatrix} = 13 \neq 0 \end{array} \right\} \Rightarrow \text{rank}(\mathbf{A}) = 2$$

Secondly, we study the rank of $\mathbf{M}$, for which we augment the 2×2 minor studied previously with the last row and columns of $\mathbf{M}$:

$$\begin{vmatrix} 3 & -1 & 1 \\ 1 & 4 & b \\ 2 & -5 & -2 \end{vmatrix} = 13b - 39. \quad |\mathbf{M}| = 0 \iff b = 3$$

We have now two sub-cases depending on the value of b

- (i) Case $a = 1, b \neq 3$

$$\left. \begin{array}{l} \text{rank}(\mathbf{A}) = 2 \\ \text{rank}(\mathbf{M}) = 3 \end{array} \right\} \Rightarrow \text{the system is inconsistent (there is no solution)}$$

- (ii) Case $a = 1, b = 3$. Here, $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 2 < n \rightarrow$ the system is consistent and dependent (it has infinitely many solutions) with one degree of freedom.
In order to find the solutions, we may choose one variable as the parameter ($z = \lambda$).
Using Cramer's rule:

$$\begin{cases} 3x - y = 1 - 2\lambda \\ x + 4y = 3 - \lambda \end{cases} \Rightarrow \begin{cases} x = \frac{\begin{vmatrix} 1 - 2\lambda & -1 \\ 3 - \lambda & 4 \end{vmatrix}}{13} = \frac{-9\lambda + 7}{13} \\ y = \frac{\begin{vmatrix} 3 & 1 - 2\lambda \\ 1 & 3 - \lambda \end{vmatrix}}{13} = \frac{-\lambda + 8}{13} \\ z = \lambda \end{cases}$$

6. The coefficient matrix of the system is $\mathbf{A} = \begin{pmatrix} 1 & -2 & a \\ 1 & 4 & a^2 \\ 1 & -8 & a^2 \end{pmatrix}$, and its determinant is

$|\mathbf{A}| = 12a^2 - 12a = 12a(a - 1)$. We notice three different cases:

- Case $a \neq 0$ and $a \neq 1$. The rank of the coefficient and augmented matrices coincide and is equal to the number of variables = 3. Following Rouché-Capelli theorem, the system is consistent and independent. We could use Cramer's rule to find the solution:

$$x = \frac{\begin{vmatrix} 1 - a & -2 & a \\ 6 & 4 & a^2 \\ -6 & -8 & a^2 \end{vmatrix}}{12a^2 - 12a} = \frac{36a^2 - 12a^3 - 24a}{12a^2 - 12a} = \frac{12a(3a - a^2 - 2)}{12a(a - 1)} = \frac{12a(a - 1)(2 - a)}{12a(a - 1)} = 2 - a$$

$$y = \frac{\begin{vmatrix} 1 & 1 - a & a \\ 1 & 6 & a^2 \\ 1 & -6 & a^2 \end{vmatrix}}{12a^2 - 12a} = \frac{12a^2 - 12a}{12a^2 - 12a} = 1$$

$$z = \frac{\begin{vmatrix} 1 & -2 & 1 - a \\ 1 & 4 & 6 \\ 1 & -8 & -6 \end{vmatrix}}{12a^2 - 12a} = \frac{-12 + 12a}{12a^2 - 12a} = \frac{12(a - 1)}{12a(a - 1)} = \frac{1}{a}$$

- Case $a = 0$. The augmented matrix is $\mathbf{M} = \left(\begin{array}{ccc|c} 1 & -2 & 0 & 1 \\ 1 & 4 & 0 & 6 \\ 1 & -8 & 0 & -6 \end{array} \right)$.

Since $\begin{vmatrix} 1 & -2 \\ 1 & 4 \end{vmatrix} = 6 \neq 0$, the rank of matrix $\mathbf{A} = 2$. On the other hand, the rank of $\mathbf{M}$ is 3, since we could find a non-zero 3x3 minor:

$$\begin{vmatrix} 1 & -2 & 1 \\ 1 & 4 & 6 \\ 1 & -8 & -6 \end{vmatrix} = -12 \neq 0.$$

Following Rouché-Capelli theorem, the system is inconsistent (i.e. it has no solution).

- Case $a = 1$. Here, the augmented matrix is $M = \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 1 & 4 & 1 & 6 \\ 1 & -8 & 1 & -6 \end{array} \right)$. The same 2x2 minor as in the previous case can be found here $\begin{vmatrix} 1 & -2 \\ 1 & 4 \end{vmatrix} = 6 \neq 0$, so $\text{rank}(A) = 2$. To compute the rank of the augmented matrix, we could look for non-zero 3x3 minors or use Gaussian elimination.

For the former option, we realise that two columns (C1 and C3) are the same so we could get rid of one of them to compute the rank of the matrix (remember that when two rows/columns are linearly dependent, the determinant is zero). Then, the only 3x3 minor is

$$\begin{vmatrix} 1 & -2 & 0 \\ 1 & 4 & 6 \\ 1 & -8 & -6 \end{vmatrix} = 0,$$

For the latter option,

$$\left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 1 & 4 & 1 & 6 \\ 1 & -8 & 1 & -6 \end{array} \right) \xrightarrow{(1)} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 6 & 0 & 6 \\ 0 & -6 & 0 & -6 \end{array} \right) \xrightarrow{(2)} \left(\begin{array}{ccc|c} 1 & -2 & 1 & 0 \\ 0 & 6 & 0 & 6 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

$$(1) R_2 = R_2 - R_1 \text{ and } R_3 = R_3 - R_1$$

$$(2) R_3 = R_3 + R_2$$

In both cases we see that $\text{rank}(M) = \text{rank}(A) = 2 < 3 = n$, where n is the number of variables in the system. Hence, the system is consistent and dependent with 1 degree of freedom.

Looking at the matrix in echelon form from the Gaussian elimination method, we see that:

$$\begin{cases} x - 2y + z = 0 \\ 6y = 6 \rightarrow y = 1 \end{cases} \Rightarrow \begin{cases} x - 2y + z = 0 \rightarrow x = 2 - z \\ y = 1 \end{cases}$$

Finally, the parameterised solution is $(x, y, z) = (2 - \lambda, 1, \lambda)$.

7. Two planes in the space could have one of the following relative positions: parallel, coincident or intersect at a line. Since we have 3 variables and 2 equations, the system could not intersect at a point (single solution) and will have either infinitely many solutions with 1 (intersect at a line) or 2 (coincident planes) degrees of freedom or would be inconsistent (parallel planes). Let us discuss the system composed of both planes

$$\begin{cases} 2x + 3y - z = -8 \\ -4x - 6y + 2z = 16 \end{cases}$$

The coefficient and augmented matrices are:

$$A = \begin{pmatrix} 2 & 3 & -1 \\ -4 & -6 & 2 \end{pmatrix} \quad M = \left(\begin{array}{ccc|c} 2 & 3 & -1 & -8 \\ -4 & -6 & 2 & 16 \end{array} \right)$$

All second order minor from matrix $\mathbf{A}$ are zero:

$$\begin{vmatrix} 2 & 3 \\ -4 & -6 \end{vmatrix} = -12 + 12 = 0, \quad \begin{vmatrix} 2 & -1 \\ -4 & 2 \end{vmatrix} = 4 - 4 = 0, \quad \begin{vmatrix} 3 & -1 \\ -6 & 2 \end{vmatrix} = 6 - 6 = 0$$

Hence, $\text{rank}(\mathbf{A}) = 1$, meaning that all rows or columns are linearly dependent of each other. Similarly, we realise that the last column of $\mathbf{M}$ is linearly dependent of any other row, so $\text{rank}(\mathbf{M}) = 1$, as well. Since both ranks coincide and are equal to $1 < 3 = n$, we know that the system is consistent and dependent with 2 degrees of freedom, i.e., both planes are coincident.

8. The line r would be parallel to π if the linear system is inconsistent (i.e., there is no solution). Using Rouché-Capelli, the condition $\text{rank}(\mathbf{A}) < \text{rank}(\mathbf{M})$ has to hold. The system of equations is:

$$\begin{cases} 4x - 4 = 2y + 10 \\ 2x - 2 = 2z - 6 \\ 4x + my + z = 2 \end{cases} \iff \begin{cases} 4x - 2y = 14 \\ 2x - 2z = -4 \\ 4x + my + z = 2 \end{cases} \iff \begin{cases} 2x - y = 7 \\ x - z = -2 \\ 4x + my + z = 2 \end{cases}$$

First, we need $|\mathbf{A}| = 0$:

$$\begin{vmatrix} 2 & -1 & 0 \\ 1 & 0 & -1 \\ 4 & m & 1 \end{vmatrix} = 0 \quad \Rightarrow \quad 4 + 2m + 1 = 0 \quad \Rightarrow \quad m = -\frac{5}{2}$$

Secondly, we need to find a non-zero 3x3 minor of matrix $\mathbf{M}$. For example:

$$\begin{vmatrix} 2 & 0 & 7 \\ 1 & -1 & -2 \\ 4 & 1 & 2 \end{vmatrix} = -4 + 7 + 28 + 4 = 35 \neq 0$$

Thereupon, for $m = -\frac{5}{2}$, π and r are parallel.

9. Each line is effectively built by a system of two equations. Hence, we could build the composed system with 4 equations and 3 variables as follows:

$$\begin{cases} 6x - 12 = 5y \\ 2y = 6z + 6 \\ 3x - 3a = 2y + 2 \\ 3y + 3 = 3z - 6 \end{cases} \iff \begin{cases} 6x - 5y = 12 \\ 2y - 6z = 6 \\ 3x - 2y = 2 + 3a \\ 3y - 3z = -9 \end{cases} \iff \begin{cases} 6x - 5y = 12 \\ y - 3z = 3 \\ 3x - 2y = 2 + 3a \\ y - z = -3 \end{cases}$$

For the lines to intersect at the space, the system needs to be consistent and independent. In other words, the condition is: $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 3 = n$.

Firstly, we need to look for a non-zero 3x3 minor from $\mathbf{A}$:

$$\begin{vmatrix} 6 & -5 & 0 \\ 0 & 1 & -3 \\ 3 & -2 & 0 \end{vmatrix} = 45 - 36 = 9 \neq 0$$

Secondly, we need that $|\mathbf{M}| = 0$:

$$\begin{vmatrix} 6 & -5 & 0 & 12 \\ 0 & 1 & -3 & 3 \\ 3 & -2 & 0 & 2+3a \\ 0 & 1 & -1 & -3 \end{vmatrix} = \frac{1}{2} \cdot \begin{vmatrix} 6 & -5 & 0 & 12 \\ 0 & 1 & -3 & 3 \\ 0 & 1 & 0 & 6a-8 \\ 0 & 1 & -1 & -3 \end{vmatrix} = \frac{1}{2} \cdot 6 \cdot \begin{vmatrix} 1 & -3 & 3 \\ 1 & 0 & 6a-8 \\ 1 & -1 & -3 \end{vmatrix} =$$

$$= 3 \cdot (-18a + 24 - 3 + 6a - 8 - 9) = 0 \Rightarrow 3 \cdot (4 - 12a) = 0 \Rightarrow a = 1/3$$

Hence, for $a = 1/3$, the matrix rank is maximum ($\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 3 = n$) and, following Rouché-Capelli theorem, the system is consistent and independent. Consequently, r and r' intersect.

10. Using matrix notation, the system reads:

$$\begin{pmatrix} 1 & -2 & 0 & 0 \\ 2 & -4 & 2 & -4 \\ -3 & 6 & 1 & -2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \\ z \\ t \end{pmatrix} = \begin{pmatrix} 3 \\ 4 \\ -10 \end{pmatrix}$$

The coefficient and augmented matrices are:

$$\mathbf{A} = \begin{pmatrix} 1 & -2 & 0 & 0 \\ 2 & -4 & 2 & -4 \\ -3 & 6 & 1 & -2 \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{cccc|c} 1 & -2 & 0 & 0 & 3 \\ 2 & -4 & 2 & -4 & 4 \\ -3 & 6 & 1 & -2 & -10 \end{array} \right)$$

We will solve the problem using two different methods.

Method 1: Gaussian elimination. We need to transform matrix $\mathbf{M}$ in echelon form. Using R_1 , R_2 and R_3 to refer to first, second and third row, respectively, the step-wise process is as follows:

- We first operate in R_2 and R_3 to place a 0 in their first column. Hence, we perform $R_2 = R_2 - 2R_1$ and $R_3 = R_3 - 3R_1$, yielding:

$$\mathbf{M} = \left(\begin{array}{cccc|c} 1 & -2 & 0 & 0 & 3 \\ 2 & -4 & 2 & -4 & 4 \\ -3 & 6 & 1 & -2 & -10 \end{array} \right) \rightarrow \left(\begin{array}{cccc|c} 1 & -2 & 0 & 0 & 3 \\ 0 & 0 & 2 & -4 & -2 \\ 0 & 0 & 1 & -2 & -1 \end{array} \right)$$

- We note that the second column is also zero. Hence, the following pivot position would be the third column. Since it is easy to operate with a value of 1 in the pivot position, we permute R_2 and R_3

$$\left(\begin{array}{cccc|c} 1 & -2 & 0 & 0 & 3 \\ 0 & 0 & 1 & -2 & -1 \\ 0 & 0 & 2 & -4 & -2 \end{array} \right)$$

- Now, we operate R_3 to get a zero in the third column. Concretely, we perform $R_3 = R_3 - 2R_2$

$$\left(\begin{array}{cccc|c} 1 & -2 & 0 & 0 & 3 \\ 0 & 0 & 1 & -2 & -1 \\ 0 & 0 & 0 & 0 & 0 \end{array} \right)$$

The updated system is the following:

$$\begin{cases} x - 2y = 3 \\ z - 2t = -1 \end{cases}$$

Hence, we have a system which is consistent and dependent, with two degrees of freedom. The overall solution is $x = 3 + 2y, y = y, z = -1 + 2t, t = t$, with y, t as parameters.

Method 2: Cramer's rule. We start by computing the rank of the coefficient matrix $\mathbf{A}$, looking at minors of all orders until we find an order with all minors that are zero.

- Starting with 1x1 minors, we clearly see that taking the first row/columns the result is $= 1 \neq 0$. From here, we can conclude that the rank of $\mathbf{A}$ is, at least, 1.
- We consider 2x2 minors that contain the previous studied 1x1 minor. The first and most obvious minor is: $\begin{vmatrix} 1 & -2 \\ 2 & -4 \end{vmatrix} = -4 + 4 = 0$. Since its determinant is zero, we need to look for another 2x2 minor. For example, taking the first and second rows and the first and third columns we get: $\begin{vmatrix} 1 & 0 \\ 2 & 2 \end{vmatrix} = 2 \neq 0$. Since its determinant is non-zero, we conclude that the rank of matrix $\mathbf{A}$ is, at least, 2.
- To check whether the matrix $\mathbf{A}$ has rank 3 or not, we need to build 3x3 minors that contain the previously analysed non-zero 2x2 minor. For example, taking first, second and third rows/columns we get:

$$\begin{vmatrix} 1 & -2 & 0 \\ 2 & -4 & 2 \\ -3 & 6 & 1 \end{vmatrix} = -4 + 12 + 4 - 12 = 0.$$

Another possible minor is using first, second and third rows with first, third and fourth columns:

$$\begin{vmatrix} 1 & 0 & 0 \\ 2 & 2 & -4 \\ -3 & 1 & -2 \end{vmatrix} = -4 + 4 = 0.$$

We have now checked all minors that contain the 2x2 minor analysed previously and their determinant is zero. Hence, we conclude that the rank of $\mathbf{A}$ is not 3 and so, $\text{rank}(\mathbf{A}) = 2$

Then, we need to check the augmented matrix:

$$\mathbf{M} = \left(\begin{array}{ccccc} 1 & -2 & 0 & 0 & 3 \\ 2 & -4 & 2 & -4 & 4 \\ -3 & 6 & 1 & -2 & -10 \end{array} \right)$$

Since the 2×2 non-zero minor found previously is also contained in matrix $\mathbf{M}$, we know that the $\text{rank}(\mathbf{M}) \geq 2$. From that minor, we look for non-zero minors of the augmented matrix. The only minor that has not been checked (recall that the two that were checked earlier are also contained in this matrix) is by gathering rows 1,2,3 and columns 1,3,5:

$$\begin{vmatrix} 1 & 0 & 3 \\ 2 & 2 & 4 \\ -3 & 1 & -10 \end{vmatrix} = -20 + 6 + 18 - 4 = 0.$$

Since the determinant is zero, we conclude that the $\text{rank}(\mathbf{M}) = \text{rank}(\mathbf{A}) = 2$. Hence, the system is consistent and dependent (i.e., there exists infinitely many solutions). The number of variables is $n = 4$ and the rank of the system is 2, yielding a total of 2 degrees of freedom. Since the 2×2 non-zero minor is built using first and third columns, we use x and z as variables and y and t as parameters. Since the minor involved first and second columns, the third equation can be removed, and the final system is as follows:

$$\begin{cases} x = 3 + 2y \\ 2x + 2z = 4 + 4y + 4t \end{cases}$$

This equivalent system is a Cramer's system with determinant $\begin{vmatrix} 1 & 0 \\ 2 & 2 \end{vmatrix} = 2$. Using Cramer's rule to find the unknown variables:

$$x = \frac{\begin{vmatrix} 3 + 2y & 0 \\ 4 + 4y + 4t & 2 \end{vmatrix}}{\begin{vmatrix} 1 & 0 \\ 2 & 2 \end{vmatrix}} = \frac{6 + 4y}{2} = 3 + 2y$$

$$z = \frac{\begin{vmatrix} 1 & 3 + 2y \\ 2 & 4 + 4y + 4t \end{vmatrix}}{\begin{vmatrix} 1 & 0 \\ 2 & 2 \end{vmatrix}} = \frac{4 + 4y + 4t - 6 - 4y}{2} = \frac{-2 + 4t}{2} = -1 + 2t$$

Finally, the overall solution is:

$$\begin{aligned} x &= 3 + 2y \\ y &= y \\ z &= -1 + 2t \\ t &= t \end{aligned}$$

11.

- a) To discuss the system we use the Rouché-Capelli theorem, for which we need to compute the rank of the coefficient and augmented matrices.

First, we check $\text{rank}(\mathbf{A})$ by computing the determinant of 3×3 minors until we get a non-zero minor:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 & k \\ 1 & 2 & 8 \\ 1 & 1 & 7 \\ 2 & 1 & k \end{pmatrix} \rightarrow \begin{vmatrix} 1 & 1 & k \\ 1 & 2 & 8 \\ 1 & 1 & 7 \end{vmatrix} = 7 - k,$$

$$\rightarrow \begin{vmatrix} 1 & 2 & 8 \\ 1 & 1 & 7 \\ 2 & 1 & k \end{vmatrix} = 13 - k.$$

Computing these two 3x3 minors we already see that they cannot be jointly zero (the first is cancelled for $k = 7$ and the second for $k = 13$). Hence, there will always be a non-zero 3x3 minor and $\text{rank}(\mathbf{A}) = 3$.

On the other hand, the augmented matrix is squared and we compute its 4x4 determinant

$$\mathbf{M} = \begin{pmatrix} 1 & 1 & k & 1 \\ 1 & 2 & 8 & k \\ 1 & 1 & 7 & 1 \\ 2 & 1 & k & 2 \end{pmatrix} \rightarrow |\mathbf{M}| = \begin{vmatrix} 1 & 1 & k & 1 \\ 1 & 2 & 8 & k \\ 1 & 1 & 7 & 1 \\ 2 & 1 & k & 2 \end{vmatrix} = -k^2 + 8k - 7.$$

Hence, $|\mathbf{M}| = 0 \iff k = \{1, 7\}$. Then, for these two values, $\text{rank}(\mathbf{M}) = 3$ while for the rest $\text{rank}(\mathbf{M}) = 4$.

Finally, we conclude that:

- For $k = \{1, 7\}$ the system is consistent and independent.
- For $k \neq \{1, 7\}$, the system is inconsistent.

b) We use Gaussian elimination to solve the system

$$\begin{pmatrix} 1 & 1 & k & 1 \\ 1 & 2 & 8 & k \\ 1 & 1 & 7 & 1 \\ 2 & 1 & k & 2 \end{pmatrix} \xrightarrow{(1)} \begin{pmatrix} 1 & 1 & k & 1 \\ 0 & 1 & 8-k & k-1 \\ 0 & 0 & 7-k & 0 \\ 0 & -1 & -k & 0 \end{pmatrix} \xrightarrow{(2)} \begin{pmatrix} 1 & 1 & k & 1 \\ 0 & 1 & 8-k & k-1 \\ 0 & -1 & -k & 0 \\ 0 & 0 & 7-k & 0 \end{pmatrix} \xrightarrow{(3)} \begin{pmatrix} 1 & 1 & k & 1 \\ 0 & 1 & 8-k & k-1 \\ 0 & 0 & 8-2k & k-1 \\ 0 & 0 & 7-k & 0 \end{pmatrix}$$

(1) $R_2 = R_2 - R_1$, $R_3 = R_3 - R_1$ and $R_4 = R_4 - 2R_1$

(2) Permutation of R_3 and R_4

(3) $R_3 = R_3 + R_2$

- For $k = 1$, we get the following system: $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 7 & 0 \\ 0 & 0 & 6 & 0 \\ 0 & 0 & 6 & 0 \end{pmatrix}$. Going bottom-up, we easily

find the solutions:

$$z = 0, \quad y = -7z = 0 \quad x = 1 - y - z = 1$$

– For $k = 7$, the augmented matrix is $\begin{pmatrix} 1 & 1 & 7 & 1 \\ 0 & 1 & 1 & 6 \\ 0 & 0 & -6 & 6 \\ 0 & 0 & 0 & 0 \end{pmatrix}$ and the solution:

$$z = -1, \quad y = 6 - z = 7 \quad x = 1 - y - 7z = 1$$

c) In the system introduced in **a)**, the first two equations correspond to line r while the other two to line s . Using the previous discussion of the system we could study the relative positions of r and s :

- For $k = 1$ or $k = 7$ we know that the system is consistent and independent thus the lines intersect at a point. Moreover, this point was actually found in **b)**: $(x, y, z) = (1, 0, 0)$ for $k = 1$ and $(x, y, z) = (1, 7, -1)$ for $k = 7$.
- For other values of $k \neq 1, 7$ the system is inconsistent and the lines do not intersect.

12. Following the Rouché-Capelli theorem, we discuss the system by comparing the rank of the coefficient and augmented matrices. Initially, we look for those parameter values that maximise the matrices' rank, in this case 3 (because we have three variables and three equations).

$$|\mathbf{A}| = \begin{vmatrix} a & b & 1 \\ 1 & ab & 1 \\ 1 & b & a \end{vmatrix} = b(a-1)^2(a+2).$$

The values that cancel the determinant are: $b = 0$, $a = \{1, -2\}$. Therefore, we can distinguish four different cases

- i)** Case $b \neq 0$ and $a \neq 1, -2$. In this case the determinant of $\mathbf{A}$ is non-zero and hence $\text{rank}(\mathbf{A}) = 3$. Since the maximum rank of the augmented matrix $\mathbf{M}$ is 3 (due to the dimensionality of the system) and the minimum is also 3 (because it contains the matrix $\mathbf{A}$, a non-zero 3×3 minor) we conclude that $\text{rank}(\mathbf{M}) = 3$ and that the system is consistent and independent, as stated by Rouché-Capelli theorem. We use Cramer's rule to compute the solution:

$$x = \frac{\begin{vmatrix} 1 & b & 1 \\ b & ab & 1 \\ 1 & b & a \end{vmatrix}}{b(a-1)^2(a+2)} = \frac{b(a-b)(a-1)}{b(a-1)^2(a+2)} = \frac{a-b}{(a-1)(a+2)}$$

$$y = \frac{\begin{vmatrix} a & 1 & 1 \\ 1 & b & 1 \\ 1 & 1 & a \end{vmatrix}}{b(a-1)^2(a+2)} = \frac{ab+b-2}{b(a-1)(a+2)}$$

$$z = \frac{\begin{vmatrix} a & b & 1 \\ 1 & ab & b \\ 1 & b & 1 \end{vmatrix}}{b(a-1)^2(a+2)} = \frac{b(a-b)(a-1)}{b(a-1)^2(a+2)} = \frac{a-b}{(a-1)(a+2)}$$

The solution of the system, depending on parameters a, b , is:

$$(x, y, z) = \left(\frac{a-b}{(a-1)(a+2)}, \frac{ab+b-2}{b(a-1)(a+2)}, \frac{a-b}{(a-1)(a+2)} \right)$$

ii) Case $b = 0$. The system matrices are:

$$\mathbf{A} = \begin{pmatrix} a & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & a \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{ccc|c} a & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & a & 1 \end{array} \right)$$

Here, the $\text{rank}(\mathbf{A}) < 3$ because $b = 0$ cancels its determinant, so we need to look for

non-zero 2x2 minors. A potential candidate is: $\begin{vmatrix} a & 1 \\ 1 & 1 \end{vmatrix} = a - 1$, and the other minors are

either identical or zero. Hence, we note two subcases:

- If $a \neq 1$, then $\text{rank}(\mathbf{A}) = 2$. On the other hand, the augmented matrix has a non-zero 3x3 minor and thus $\text{rank}(\mathbf{M}) = 3$.

$$\begin{vmatrix} a & 1 & 1 \\ 1 & 1 & 0 \\ 1 & a & 1 \end{vmatrix} = 2(a-1) \neq 0$$

Following Rouché-Capelli theorem, the system is inconsistent.

- If $a = 1$, the matrices are:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{ccc|c} 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{array} \right)$$

We observe that $\text{rank}(\mathbf{A}) = 1$ and $\text{rank}(\mathbf{M}) = 2$ and conclude that, in this case, the system is also inconsistent.

iii) Case $a = 1$. In this case, the system matrices are:

$$\mathbf{A} = \begin{pmatrix} 1 & b & 1 \\ 1 & b & 1 \\ 1 & b & 1 \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{ccc|c} 1 & b & 1 & 1 \\ 1 & b & 1 & b \\ 1 & b & 1 & 1 \end{array} \right).$$

We observe that $\text{rank}(\mathbf{A}) = 1$ regardless of the value of b because all rows and columns are linearly dependent between each other. Regarding the augmented matrix, all 3x3

minors are zero and its rank depends on the value of b : $\begin{vmatrix} 1 & 1 \\ 1 & b \end{vmatrix} = b - 1$. Again, we note

two different subcases:

- If $b \neq 1$, then $\text{rank}(\mathbf{M}) = 2$ and the system is inconsistent.
- If $b = 1$, then $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 1$ and the system is consistent and dependent, with two degrees of freedom. Therefore, the system has one equation and three variables:

$x + y + z = 1$. Using y, z as parameters, the solution of this system is:

$$(x, y, z) = (1 - y - z, y, z) \quad \forall y, z \in \mathbb{R}$$

iv) Case $a = -20$. In this last case, the system matrices are as follows:

$$\mathbf{A} = \begin{pmatrix} -2 & b & 1 \\ 1 & -2b & 1 \\ 1 & b & -2 \end{pmatrix} \quad \mathbf{M} = \left(\begin{array}{ccc|c} -2 & b & 1 & 1 \\ 1 & -2b & 1 & b \\ 1 & b & -2 & 1 \end{array} \right).$$

We know that the $\text{rank}(\mathbf{A}) = 2$ because there is at least a 2×2 non-zero minor $\begin{vmatrix} 1 & 1 \\ 1 & -2 \end{vmatrix} = -3$, and that the $\text{rank}(\mathbf{M})$ depends on parameter b value:

$$\begin{vmatrix} -2 & 1 & 1 \\ 1 & 1 & b \\ 1 & -2 & 1 \end{vmatrix} = -6 - 3b.$$

Two subcases are observed:

- If $b \neq -2$, then $\text{rank}(\mathbf{M}) = 3$ and thus the system is inconsistent.
- If $b = -2$, then $\text{rank}(\mathbf{M}) = 2$ and thus the system is consistent and dependent with one degree of freedom. We use only two of the three equations and z as the free parameter:

$$\begin{cases} x + 4y + z = -2 \\ x - 2y - 2z = 1 \end{cases} \rightarrow \begin{cases} x + 4y = -2 - z \\ x - 2y = 1 + 2z \end{cases}$$

Using matrix notation,

$$\begin{pmatrix} 1 & 4 \\ 1 & -2 \end{pmatrix} \cdot \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} -2 - z \\ 1 + 2z \end{pmatrix}$$

Following Cramer's rule

$$x = \frac{\begin{vmatrix} -2 - z & 4 \\ 1 + 2z & -2 \end{vmatrix}}{\begin{vmatrix} 1 & 4 \\ 1 & -2 \end{vmatrix}} = \frac{-6z}{-6} = z,$$

$$y = \frac{\begin{vmatrix} 1 & -2 - z \\ 1 & 1 + 2z \end{vmatrix}}{\begin{vmatrix} 1 & 4 \\ 1 & -2 \end{vmatrix}} = \frac{1 + 2z + 2 + z}{-6} = \frac{3z + 3}{-6} = \frac{z + 1}{-2},$$

the solution of the system is $(x, y, z) = (z, \frac{-1-z}{2}, z)$ for any $z \in \mathbb{R}$.

On the whole, the relative position of the three planes is:

- If $b \neq 0$ and $a \neq \{1, -2\}$, the system is consistent and independent and the three planes intersect at a single point. This point is $\left(\frac{a-b}{(a-1)(a+2)}, \frac{ab+b-2}{b(a-1)(a+2)}, \frac{a-b}{(a-1)(a+2)}\right)$
- If $b = 0$ and $a \neq 1$, the system is inconsistent and the three planes do not intersect.
- If $b = 0$ and $a = 1$, the system is inconsistent and the three planes do not intersect.
- If $a = 1$ and $b \neq 1$, the system is inconsistent and the three planes do not intersect.
- If $a = 1$ and $b = 1$, the system is consistent and dependent with 2 degrees of freedom and thus, the three planes are coincident.
- If $a = -2$ and $b \neq -2$, the system is inconsistent and the three planes do not intersect.
- If $a = -2$ and $b = -2$, the system is consistent and dependent with 1 degree of freedom and thus the three planes intersect at a line parameterised as $(z, \frac{-1-z}{2}, z)$ for any $z \in \mathbb{R}$.

13. Using the equilibrium condition on every node (incoming data flow = outgoing data flow) we could write the following SLE:

$$\begin{cases} x_1 + x_4 + x_5 = 500 \\ x_1 + x_2 = 200 \\ x_4 + x_3 - x_2 = 400 \\ x_3 - x_5 = 100 \end{cases}.$$

The coefficient and augmented matrices of the system are:

$$\mathbf{A} = \begin{pmatrix} 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & -1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & -1 \end{pmatrix}, \quad \mathbf{M} = \left(\begin{array}{ccccc|c} 1 & 0 & 0 & 1 & 1 & 500 \\ 1 & 1 & 0 & 0 & 0 & 200 \\ 0 & -1 & 1 & 1 & 0 & 400 \\ 0 & 0 & 1 & 0 & -1 & 100 \end{array} \right).$$

Computing their rank as in previous exercises, we note that $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 3$. Since the number of variables is 5, the system is consistent and dependent, with two degrees of freedom. The final solution is:

$$(x_1, x_2, x_3, x_4, x_5) = (500 - x_4 - x_5, x_4 + x_5 - 300, 100 + x_5, x_4, x_5)$$

Hence, if we fix the traffic in $x_4 = 100$ and $x_5 = 200$, the overall network traffic would be:

$$\begin{cases} x_1 = 200 \\ x_2 = 0 \\ x_3 = 300 \\ x_4 = 100 \\ x_5 = 200 \end{cases}.$$

14. We define A, B, C, D variables as follows:

A = total number of A services
 B = total number of B services
 C = total number of C services
 D = total number of D services

From the total number of resources available and the resources needed for each type of service, we could build the following SLE

$$\begin{cases} A + 10B + 0C + 10D = 152 \\ 2A + B + 4C + 0D = 550 \\ 2A + B + C + D = 309 \\ 0A + B + 0C + D = 4 \end{cases},$$

with 4 equations and 4 variables that could be solved using Gaussian elimination:

$$\begin{pmatrix} 1 & 10 & 0 & 10 & 152 \\ 2 & 1 & 4 & 0 & 550 \\ 2 & 1 & 1 & 1 & 309 \\ 0 & 1 & 0 & 1 & 4 \end{pmatrix} \xrightarrow{(1)} \begin{pmatrix} 1 & 10 & 0 & 10 & 152 \\ 0 & 1 & 0 & 1 & 4 \\ 2 & 1 & 4 & 0 & 550 \\ 2 & 1 & 1 & 1 & 309 \end{pmatrix} \xrightarrow{(2)} \begin{pmatrix} 1 & 10 & 0 & 10 & 152 \\ 0 & 1 & 0 & 1 & 4 \\ 0 & -19 & 4 & -20 & 246 \\ 0 & -19 & 1 & -19 & 5 \end{pmatrix} \xrightarrow{(3)} \\ \xrightarrow{(3)} \begin{pmatrix} 1 & 10 & 0 & 10 & 152 \\ 0 & 1 & 0 & 1 & 4 \\ 0 & 0 & 4 & -1 & 322 \\ 0 & 0 & 1 & 0 & 81 \end{pmatrix} \xrightarrow{(4)} \begin{pmatrix} 1 & 10 & 0 & 10 & 152 \\ 0 & 1 & 0 & 1 & 4 \\ 0 & 0 & 4 & -1 & 322 \\ 0 & 0 & 0 & 1 & 2 \end{pmatrix}$$

(1) Move R_4 up to R_2 and bring R_3 and R_4 downwards.

(2) $R_3 = R_3 - 2R_1$ and $R_4 = R_4 - 2R_1$

(3) $R_3 = R_3 + 19R_2$ and $R_4 = R_4 + 19R_2$

(4) $R_4 = 4R_4 - R_3$

From the last matrix, we get that $A = 112, B = 2, C = 81, D = 2$

15.

a) We define the system variables x = as the number of photographs of normal quality and y = as the number of photographs of optimal quality. The proposed system is:

$$\begin{cases} x + y = 24 \\ 0.2x + ay = 9.2 \end{cases}.$$

Firstly, we need to examine the number of solutions studying the rank of the system matrices:

$$\mathbf{A} = \begin{pmatrix} 1 & 1 \\ 0.2 & a \end{pmatrix} \rightarrow |\mathbf{A}| = a - 0.2,$$

$$\mathbf{M} = \left(\begin{array}{cc|c} 1 & 1 & 24 \\ 0.2 & a & 9.2 \end{array} \right) \rightarrow \left| \begin{array}{cc} 1 & 24 \\ 0.2 & 9.2 \end{array} \right| = 4.4 \neq 0,$$

yielding two different cases:

- Case $a = 0.2$. The system is inconsistent because $\text{rank}(\mathbf{A}) = 1$ and $\text{rank}(\mathbf{M}) = 2$.

- Case $a \neq 0.2$. The system is consistent and independent because $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 2 = n$, where n is the number of variables. Using Cramer's rule::

$$x = \frac{\begin{vmatrix} 24 & 1 \\ 9.2 & a \end{vmatrix}}{a - 0.2} = \frac{24a - 9.2}{a - 0.2},$$

$$y = \frac{\begin{vmatrix} 1 & 24 \\ 0.2 & 9.2 \end{vmatrix}}{a - 0.2} = \frac{4.4}{a - 0.2},$$

we find the following solution:

- The number of photographs of normal quality is $x = \frac{24a - 9.2}{a - 0.2}$
 - The number of photographs of optimal quality is $y = \frac{4.4}{a - 0.2}$
- b)** Yes there is. Following the results from **a)**, if $a = 0.2$ MB it is impossible to find the number of pictures of optimal quality, since the denominator of $\frac{4.4}{a - 0.2}$ is 0 (you can also argue that the system is inconsistent).
- c)** We add an extra constraint to the SLE such that $x = y$. From **a)**,

$$x = y \rightarrow \frac{24a - 9.2}{a - 0.2} = \frac{4.4}{a - 0.2} \rightarrow 24a - 9.2 = 4.4 \rightarrow 24a = 13.6 \rightarrow a = 0.56 \text{ MB}$$

16. The system variables are the number of disks of type A (x) and B (y) acquired by the system administrator. Then, the optimisation problem can be written as:

$$\begin{aligned} &\text{maximise } f(x, y) = 300x + 200y \\ &\text{s.t } 200x + 100y \leq 1500 \quad \text{price restriction} \\ &\quad x + 2y \leq 27 \quad \text{time restriction} \\ &\quad 15x + 3y \leq 100 \quad \text{energy restriction} \\ &\quad x \geq 0 \quad \text{disk A quantity can't be negative} \\ &\quad y \geq 0 \quad \text{disk B quantity can't be negative.} \end{aligned}$$

The goal of the system administrator is to maximise the memory available subject to (s.t.) different constraints. The last two conditions are intrinsic to the problem, since the quantity of disks cannot be zero, while the other conditions are given by the exercise. This type of problems are a special case of the more broader field of mathematical optimisation called *linear programming*. Different well-known algorithms could be used to compute the solution (e.g., the simplex method). Fortunately, the vast majority of mathematical software include related functions that automate the calculations.

Using our favourite software, we find the solution of the problem: $x = 1$ disk of type A and $y = 13$ disks of type B. Hence, the memory available in the LAN is $f(x, y) = 300x + 200y = 300 \cdot 1 + 200 \cdot 13 = 2900$ MB.

17.

- a)** The determinant of matrix $\mathbf{A}$ is:

$$\det(\mathbf{A}) = -3359232k^3 - 16796160k^2 - 26873856k - 13436928$$

- b)** The $\text{rank}(\mathbf{A})$ is not maximum for those k values that cancel the determinant: $k = \{-1, -2\}$, i.e., the roots of the polynomial found in **a)**.

- c) From previous results we know that for $k \neq \{-1, -2\}$ the $\text{rank}(\mathbf{A}) = 7$. With the help of mathematical software, it's easy to check that for $k = -1$ then $\text{rank}(\mathbf{A}) = 6$ while for $k = -2$ then $\text{rank}(\mathbf{A}) = 5$.
- d) The augmented matrix of the system is as follows:

$$\mathbf{M}(k) = \mathbf{A}(k)|\mathbf{B} = \left(\begin{array}{ccccccc|c} 6 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 9 & 12k-15 & -12k+3 & 15 & 12k-3 & 0 & -9 & 0 \\ 2k-13 & 2k-5 & 2k+9 & -1 & 4k+3 & 0 & -13 & 0 \\ 30 & 12k-18 & -12k+18 & 18 & 12k-18 & 0 & 6 & 0 \\ 2k-16 & -4k+16 & 8k & -16 & -2k+12 & 0 & -4 & 0 \\ 22k-5 & -14k+11 & 16k-3 & 6k-5 & -16k+3 & 6k+12 & -5 & -4 \\ 0 & -6k+6 & 6k-6 & 0 & -6k+6 & 0 & 12 & 0 \end{array} \right).$$

For $k \neq \{-1, -2\}$ the coefficient and augmented matrix have maximum rank. The relationship of matrices' rank for the other cases is:

$$\begin{array}{ll} \text{rank}(\mathbf{A}(k = -1)) = 6, & \text{rank}(\mathbf{A}(k = -2)) = 5 \\ \text{rank}(\mathbf{M}(k = -1)) = 6, & \text{rank}(\mathbf{M}(k = -2)) = 6 \end{array}$$

Consequently,

- If $k \neq \{-1, -2\}$, then the system is consistent and independent, as $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 7 = n$
- If $k = -1$, then the system is consistent and dependent with one degree of freedom, as $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{M}) = 6 < n$
- If $k = -2$, then the system is inconsistent, as $\text{rank}(\mathbf{A}) = 5 < \text{rank}(\mathbf{M})$

18.

- a) The matrices associated with the system are:

$$\mathbf{A}(k) = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ k & 10 & 4 \end{pmatrix}, \quad \mathbf{M}(k) = \left(\begin{array}{ccc|c} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ k & 10 & 4 & 11 \end{array} \right),$$

where the (k) between brackets only indicates that the matrix depends on parameter k .

The three planes intersect at a line when the system is consistent and dependent, with one degree of freedom; thus, the following condition needs to hold:

$$\text{rank}(\mathbf{A}(k)) = \text{rank}(\mathbf{M}(k)) = 2 = n - 1 \xrightarrow{\text{which holds if and only if}} k = 7$$

Solving the system for $k = 7$ using Gaussian elimination:

$$\begin{pmatrix} 1 & 1 & 1 & 2 \\ 2 & 3 & 1 & 3 \\ 7 & 10 & 4 & 11 \end{pmatrix} \xrightarrow{(1)} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 3 & -3 & -3 \end{pmatrix} \xrightarrow{(2)} \begin{pmatrix} 1 & 1 & 1 & 2 \\ 0 & 1 & -1 & -1 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$$(1) R_2 = R_2 - 2R_1 \text{ and } R_3 = R_3 - 7R_1$$

$$(2) R_3 = R_3 - 3R_2$$

The systems associated to all matrices are equivalent, so we could easily compute the solution using the latter matrix:

$$z = z;$$

$$y = -1 + z;$$

$$x = 3 - 2z.$$

- b)** In order to intersect at a point, the system build using the three planes needs to be consistent and independent therefore $\text{rank}(\mathbf{A}(k)) = \text{rank}(\mathbf{M}(k)) = 3$. This is achieved using $k \neq 7$.

For $k = 10$, we could use Cramer's rule to obtain the coordinates of the intersection point:

$$x = \frac{\begin{vmatrix} 2 & 1 & 1 \\ 3 & 3 & 1 \\ 11 & 10 & 4 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 10 & 10 & 4 \end{vmatrix}} = 0, \quad y = \frac{\begin{vmatrix} 1 & 2 & 1 \\ 2 & 3 & 1 \\ 10 & 11 & 4 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 10 & 10 & 4 \end{vmatrix}} = \frac{1}{2}, \quad z = \frac{\begin{vmatrix} 1 & 1 & 2 \\ 2 & 3 & 3 \\ 10 & 10 & 11 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ 2 & 3 & 1 \\ 10 & 10 & 4 \end{vmatrix}} = \frac{3}{2}.$$

19.

- a)** The matrices associated to the system are:

$$\mathbf{A}(k) = \begin{pmatrix} 3 & k & k \\ 1 & -1 & 0 \\ 3 & -2 & 0 \end{pmatrix}, \quad \mathbf{M}(k) = \left(\begin{array}{ccc|c} 3 & k & k & 0 \\ 1 & -1 & 0 & 0 \\ 3 & -2 & 0 & 0 \end{array} \right).$$

Since the system is homogeneous (all elements of the constant vector are zero and thus there exists the trivial solution of all zeros), we know that $\text{rank}(\mathbf{A}(k)) = \text{rank}(\mathbf{M}(k))$ and thus, the system is always consistent.

Noting that $\text{rank}(\mathbf{A}(k)) \geq 2$ since $\begin{vmatrix} 1 & -1 \\ 3 & -2 \end{vmatrix} \neq 0$, and that $|\mathbf{A}(k)| = -2k + 3k = k$, the system is:

- Independent for $k \neq 0$, since $\text{rank}(\mathbf{A}(k)) = \text{rank}(\mathbf{M}(k)) = 3 = n$
- Dependent with 1 degree of freedom for $k = 0$, since $\text{rank}(\mathbf{A}(0)) = \text{rank}(\mathbf{M}(0)) = 2 < n$

b) For $k = 1$, we solve the system using Gaussian elimination as follows:

$$\mathbf{M}(1) = \begin{pmatrix} 3 & 1 & 1 & 0 \\ 1 & -1 & 0 & 0 \\ 3 & -2 & 0 & 0 \end{pmatrix} \xrightarrow{(1)} \begin{pmatrix} 1 & -1 & 0 & 0 \\ 3 & 1 & 1 & 0 \\ 3 & -2 & 0 & 0 \end{pmatrix} \xrightarrow{(2)} \begin{pmatrix} 1 & -1 & 0 & 0 \\ 0 & 4 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix}$$

(1) Permutation of R_1 and R_2

(2) $R_2 = R_2 - 3R_1$ and $R_3 = R_3 - 3R_1$

From the last matrix we get an equivalent system whose solution is straightforward:

$$x = y = z = 0$$

Note: we already knew that for $k \neq 0$ the system is consistent and independent whose single solution is the trivial; to admit other solutions we need a dependent system ($k = 0$).

20.

a) The coefficient and augmented matrices of the system are:

$$\mathbf{A} = \begin{pmatrix} 1 & -3 & 1 \\ 3 & 6 & 1 \\ 2 & -1 & 1 \end{pmatrix}, \quad \mathbf{M} = \left(\begin{array}{ccc|c} 1 & -3 & 1 & 0 \\ 3 & 6 & 1 & 3 \\ 2 & -1 & 1 & 1 \end{array} \right).$$

The determinant is easily computed: $|\mathbf{A}| = 6 - 3 - 6 - 12 + 1 + 9 = -5$. Using Cramer's rule:

$$x = \frac{\begin{vmatrix} 0 & -3 & 1 \\ 3 & 6 & 1 \\ 1 & -1 & 1 \end{vmatrix}}{-5} = \frac{3}{5}, \quad y = \frac{\begin{vmatrix} 1 & 0 & 1 \\ 3 & 3 & 1 \\ 2 & 1 & 1 \end{vmatrix}}{-5} = \frac{1}{5}, \quad z = \frac{\begin{vmatrix} 1 & -3 & 0 \\ 3 & 6 & 3 \\ 2 & -1 & 1 \end{vmatrix}}{-5} = 0.$$

b) Following the results from **a)**, we know that the system is consistent and independent therefore all planes intersect at a single point, which is $(x, y, z) = (3/5, 1/5, 0)$.

Glossary

Augmented matrix of a system Matrix associated to a linear system that is build concatenating the coefficient matrix with the constant vector.

Coefficient matrix of a system Matrix associated to a linear system build using the coefficients of each variable $x_1, x_2, \dots, x_n$ in every equation. Each row corresponds to an equation while each column corresponds to a variable.

Consistent system System of equations that has at least one solution. Systems with no solution are called inconsistent.

Cramer's rule A closed-form expression for each entry, in the form of a quocient of determinants, that gives values to variables in a linear system of n equations, n variables that is consistent and independent.

Gaussian elimination algorithm A systematic method using elementary row operations that reduce a matrix (associated to a linear system of equations) to echelon form.

Homogeneous system System of equations whose constant vector is the zero vector.

Independent system System that has a single solution. Systems with multiple solutions are called dependent.

Linear system Set of first order equations with n variables.

Pivot position Non-zero element from a matrix that is chosen to reduce to zero the remaining coefficients in a given step of the Gaussian elimination algorithm.

Rouché-Capelli theorem Theorem that studies the consistency of a given system of equations. For consistent systems, it indicates whether a system is independent or dependent and, in this case, the degrees of freedom of the general solution

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